BINARY BEACH
Engineering & Software · Honolulu, Hawai'i

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ENGINEERING DESIGN & TEST REPORT REFERENCE DESIGN

Tradewind TW-1 · Solar LoRaWAN field sensor

Rev B · BB-ENG-2026-014 · October 12, 2026 · Hardware & Product Design · Concept to field pilot

Rev B meets 13 of 15 requirements outright, one marginally, and one (IP66 jet) is open pending an accredited lab. It sleeps at 7.2 µA, averages 76 µA at a 5-minute interval, and kept its battery above 95 % through a 30-day pilot in the Hawai'i sun.



Tradewind TW-1 as installed: ventilated solar roof, sealed electronics box, radiation shield for the air probe, pole bracket. Rendered from the production CAD.

7.2 µA
Sleep current, whole board (spec ≤ 10)

13.2 km
Range decoded at SF10, line of sight (spec ≥ 5)

37.1 °C
Hottest cell in the field, air at 31.9 °C

±0.20 °C
Worst air-temperature error, −10 to +60 °C

99.3 %
Uplinks received, 3 units × 30 days

$16.43
Electronics per unit at 1,000 (PCB + assembly incl.)

Prepared by Maxwell Pauly, Binary Beach LLC. Design files, firmware, simulations and test logs: github.com/binaryBeachLLC/tradewind. Bench and field figures in this edition are representative values for the reference design; the analysis scripts run unchanged on new logs.

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1 · SUMMARY

The product. A sealed, solar-powered sensor that reports air temperature and humidity, soil moisture and rainfall over LoRaWAN every 5 minutes (1 minute in rapid mode), for farms and remote sites. No buttons, no battery changes: a magnet tap on the lid wakes it, and the panel keeps a protected 18650 cell topped up.

The hard part was heat, not electrons. Li-ion cells must not charge above 45–50 °C, and a box in the Hawaiian sun gets far hotter than the air. Simulation showed a grey box reaching 51 °C and a panel glued to the lid 55 °C (5.4 h a day with charging blocked). Rev B mounts the panel on a ventilated roof 25 mm above a white ASA box: 35 °C in simulation, 37.1 °C measured in September.

What changed from rev A. Bring-up found two bugs: sleep current of 34 µA (the battery-sense divider leaked into the ADC clamp diode) and 2–6 rain counts per bucket tip (reed bounce). Both were reproduced in simulation, fixed in rev B and verified. The antenna match was tuned with the lid closed, which moved return loss at 915 MHz from -11.1 to -32.7 dB.

How the work ran

PHASE	WHAT HAPPENED	OUTPUT (IN THE REPO)	TYPICAL
1 · Concept	Use cases, 15 requirements, power and thermal budgets, architecture	docs/design-spec.md	1 wk
2 · Simulation	10 models: solar, thermal, energy year, power integrity, battery sense, debounce, RF match, trace impedance, link budget, board SI/PI from the routed layout	sim/	1 wk
3 · Schematic + PCB	KiCad 9, 4-layer, 85 parts; generated from one design file, ERC/DRC clean	hardware/	2 wk
4 · Enclosure	Parametric CAD fit-checked against the board STEP; drawings, STL, STEP	mechanical/	1 wk
5 · Rev A	3 boards built and brought up; 2 bugs found, root-caused, simulated, fixed	test/BRINGUP.md	3 wk
6 · Rev B	10 boards; bench and chamber characterization; production test	test/	3 wk
7 · Qualification	Immersion, ESD, thermal cycling, UV, drop; 30-day field pilot	test/	5 wk
8 · Release	Fab package, BOM, datasheet, this report, renders	docs/, render/	1 wk

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REQUIREMENTS COMPLIANCE

Every requirement from the design specification, measured on rev B. Methods and raw logs: sections 9–10 and `test/`.

ID	REQUIREMENT	TARGET	MEASURED (REV B)	RESULT
R1	Uplink interval 5 min default, 1 min rapid	5 min / 1 min	5-min cadence held for 30 days (25739 of 25920 uplinks received); 1-min rapid mode run 14 days (dark test)	PASS
R2	Sleep current, whole board at 3.7 V	≤ 10 µA	7.2 µA mean, 8.1 µA max (10 boards)	PASS
R3	Average current, 5-min interval, SF9	≤ 100 µA	75.9 µA (PPK2, sim 78 µA)	PASS
R4	Dark autonomy, 1-min rapid mode	≥ 60 days	251 days to a 20 % reserve (projected from a 14-day dark test, 0.32 %/day)	PASS
R5	Energy-positive every month incl. 7-day overcast	yes	Sept field: cell held ≥ 94.6 % with the charger in top-off standby; clear day 2336 mAh in, cloudy day 862 mAh in vs ≈ 12 mAh/day used (1-min)	PASS
R6	Charging only inside 0–50 °C	0–50 °C	stops at 50.4 °C rising / resumes 47.8 °C; stops at -0.6 °C falling (cold)	PASS · MARGINAL
R7	Air temperature accuracy, external probe	±0.3 °C	0.20 °C worst of 5 units, −10 to +60 °C (RMS 0.10 °C)	PASS
R8	Humidity accuracy, external probe, 10–90 %RH	±2 %RH	0.9 %RH worst below 90 %RH; 1.6 %RH at 97 %RH	PASS
R9	Radio range, line of sight, SF10	≥ 5 km	decoded at 13.2 km (last stop on the test road); SF7 ≥ 50 % delivery to 8.8 km	PASS
R10	Ingress protection	IP66 (target), 1 m dunk stretch	30 min at 1 m: no ingress (dew-point drift -0.05 °C); IPX5-style hose test passed	OPEN
R11	Operating temperature	−10 to +60 °C	uplinks, sensors and sleep current OK at every chamber step −10 to +60 °C; 50 thermal cycles passed	PASS
R12	ESD on external connectors	±8 kV contact	±8 kV contact: criterion A on all ports; ±15 kV air: one self-recovering reset on J4 (criterion B)	PASS
R13	Conducted TX power at U.FL, 902.3–914.9 MHz	+20 dBm ±1 dB	19.46 to 20.46 dBm, mean 19.93 dBm (2 units × 64 channels)	PASS
R15	Rain counting: no double counts, counts while asleep	exact	1199 of 1200 bench tips counted (exact up to 5 tips/s, gauge max ≈ 2/s); field vs manual gauge slope 0.988	PASS
R14	BOM cost, electronics, 1k units	≤ $25	$16.43 (parts $13.83 + PCB + assembly)	PASS

Open: R10 needs the 100 L/min IP66 jet at an accredited lab (immersion and hose already pass). **Marginal:** R6 charge-window trip points sit 0.4–0.6 °C outside nominal, inside the charger IC's tolerance.

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2 · SYSTEM ARCHITECTURE

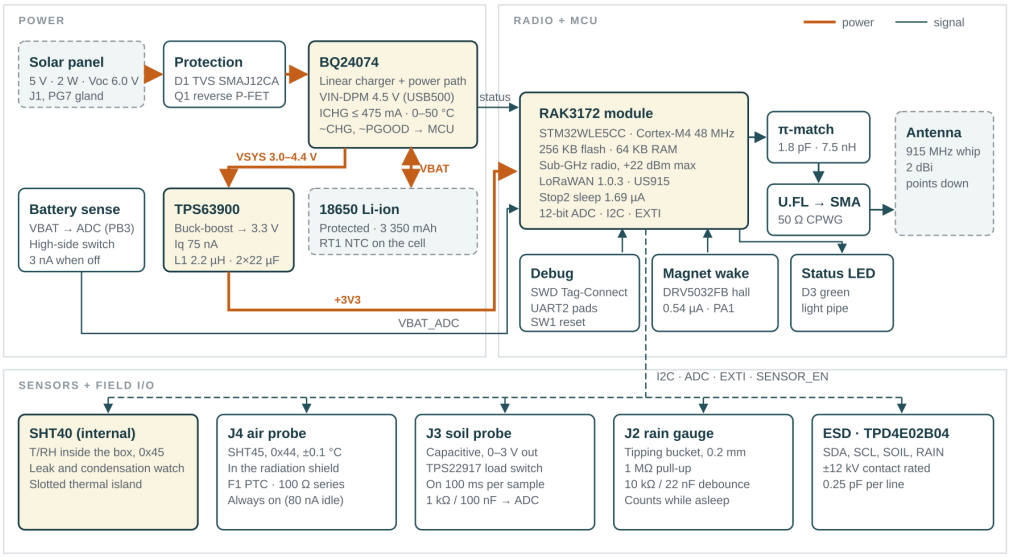


Figure 1. Functional block diagram. Power path in orange: panel → protection → BQ24074 charger/power-path → TPS63900 buck-boost → 3.3 V bus.

Power tree

RAIL	SOURCE	RANGE
VIN_SOLAR	Panel via D1, Q1	0–6.0 V
VSYS	BQ24074 OUT (power path)	VBAT … 4.4 V
VBAT	Protected 18650	3.0–4.2 V
+3V3	TPS63900 buck-boost	3.30 V ±1.5 %
+3V3_SW	TPS22917 (soil probe)	on 100 ms/sample
+3V3_EXT	F1 PTC (air probe)	always on

Key decisions

DECISION	WHY
RAK3172 module	Pre-certified radio (FCC modular), STM32WL in one part, 1.7 µA sleep
BQ24074, USB500 mode	VIN-DPM holds the panel at 4.5 V: 99 % of MPP when hot, no MPPT chip
Buck-boost, not LDO	Cell crosses 3.3 V; 75 nA Iq; 94 % at 10 mA
Ventilated solar roof	Keeps the cell under 45 °C; panel doubles as sunshade
Magnet wake	No holes in the box; 0.54 µA hall switch

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3 · DESIGN ANALYSIS AND SIMULATION

Every model is a runnable script in `sim/` (Python + ngspice); `sim/run_all.sh` regenerates every figure in about 20 s.

3.1 Solar input: fixed-voltage tracking is enough

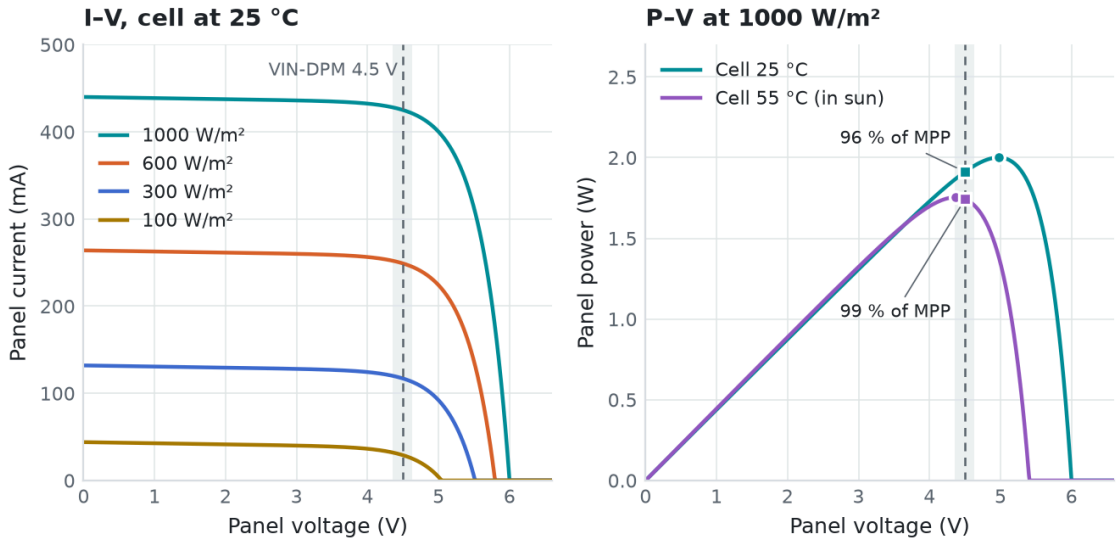


Figure 2. Single-diode model of the 5 V / 2 W panel (10 cells) at four irradiance levels, with the BQ24074 VIN-DPM window (4.35–4.63 V). At 1000 W/m² the fixed 4.5 V point captures 95.6 % of maximum power at 25 °C and 99.3 % with the panel at 55 °C, its real operating temperature.

3.2 Thermal: the battery limits the design

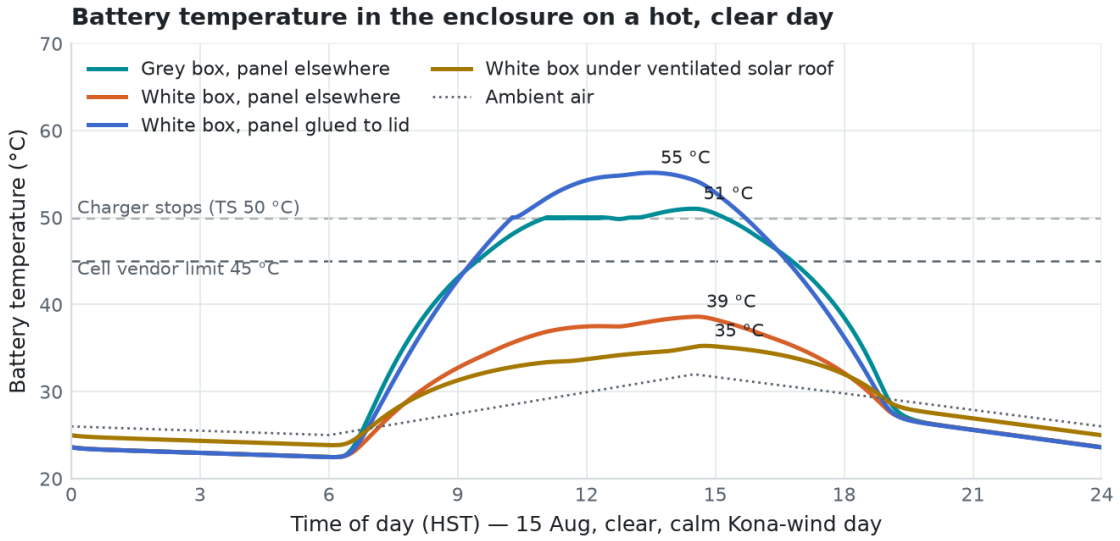


Figure 3. Lumped thermal model of the enclosure on a calm, clear August day (worst case). The roof keeps the cell at 35.3 °C; a grey box reaches 51 °C and blocks charging 2.8 h; a panel glued to the lid reaches 55 °C and blocks charging 5.4 h a day.

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3.3 Energy over a year

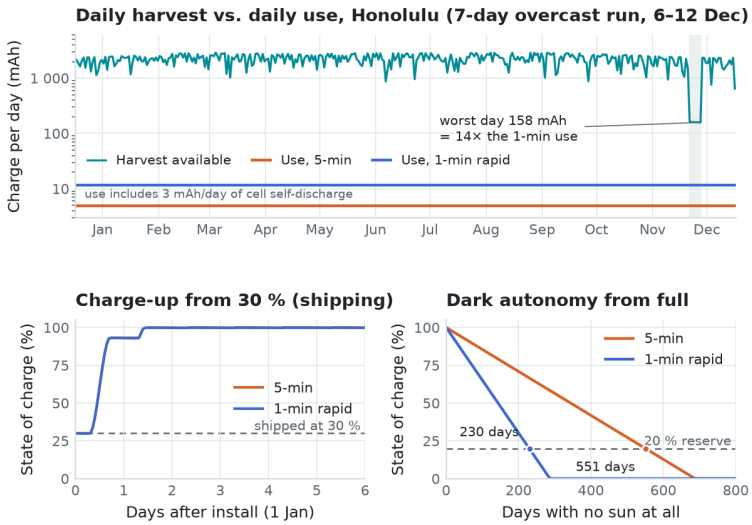


Figure 4. Hourly simulation of a Honolulu year (clear-sky model × monthly clearness, random cloud, a forced 7-day overcast run in the worst month), with charging blocked whenever the thermal model puts the cell outside 0–50 °C. Median harvest 2,324 mAh/day; the worst day still brings 158 mAh, 14× the 1-minute-mode use.

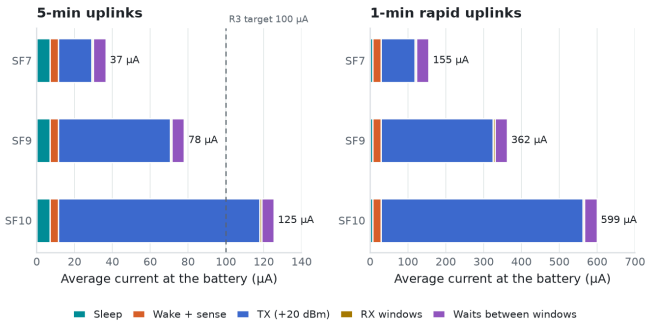


Figure 5. Where each uplink's charge goes (5-min, SF9). Transmit dominates; sleep is 9 % of the day.

MODE	AVG CURRENT	MAH/DAY	DARK DAYS*
5 min, SF7	37 µA	0.88	691
5 min, SF9	**78 µA**	1.87	550
5 min, SF10	125 µA	3.01	446
1 min, SF9	362 µA	8.69	229
1 min, SF10	599 µA	14.38	154

*No sun at all, 80 % of 3 350 mAh usable, incl. Li-ion self-discharge. Sleep 6.9 µA: charger BAT leakage 4.3 µA is the largest single term.

The 2 W panel is about 100× larger than the energy need. It is sized as the sunshade, not as an energy source; a 0.5 W panel would also be energy-positive but would not shade the box.

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3.4 Power integrity during transmit

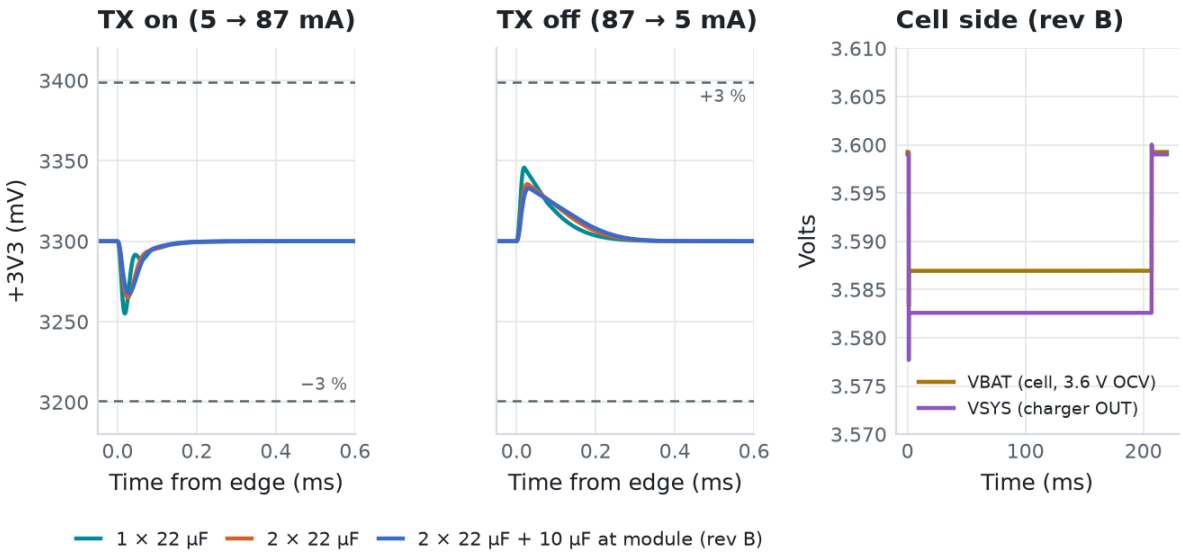


Figure 6. ngspice: cell (with internal and holder resistance) → power path → behavioural TPS63900 → +3V3, with the 87 mA TX step. Rev B's 2 × 22 µF + 10 µF keeps droop to 33 mV (spec ±99 mV); the cell sags 12 mV.

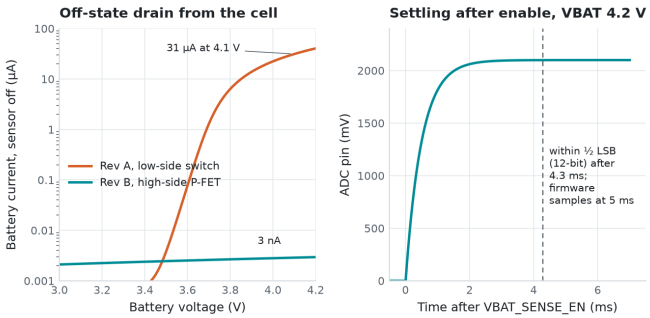


Figure 7. Battery-sense switch. Rev A (low-side) leaks 27 µA at 4.05 V and 40 µA at 4.2 V through the ADC clamp; rev B (high-side P-FET) leaks 3 nA. Settles in 4.3 ms, so firmware samples at 5 ms.

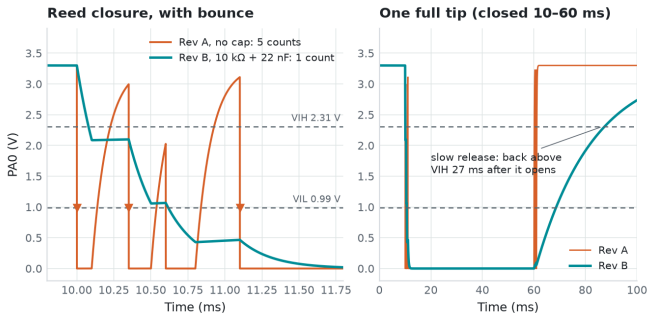


Figure 8. Reed bounce on the rain input. Rev A counted 5 edges per tip; rev B's 10 kΩ / 22 nF filter gives exactly 1. Supports 13 tips/s vs about 2 in a downpour.

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3.5 RF: matching, trace impedance and link budget

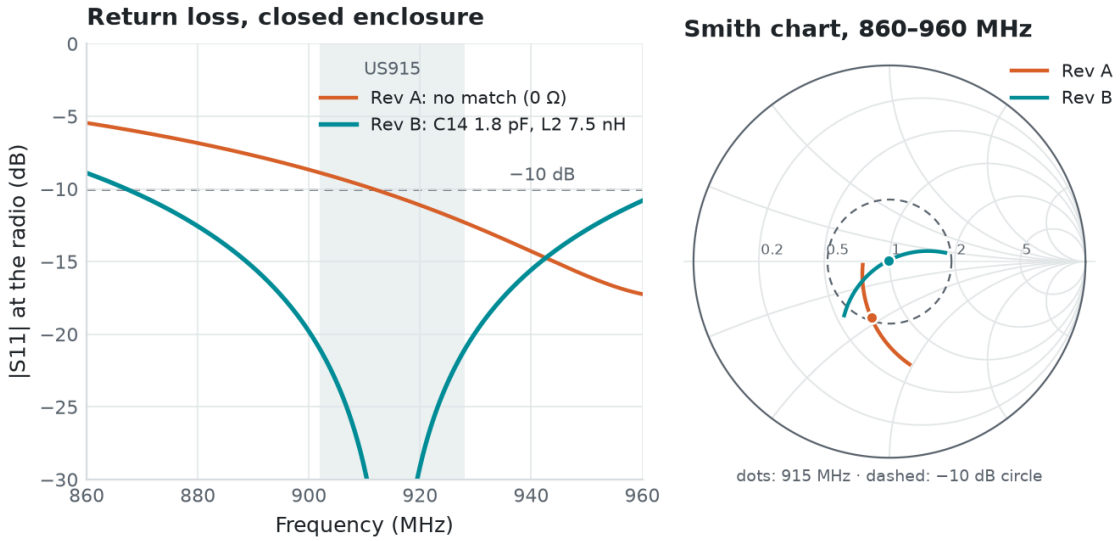


Figure 9. The whip, its 150 mm pigtail and the plastic case detune the antenna to about 38 − j21 Ω at 915 MHz. A shunt 1.8 pF at the module plus a series 7.5 nH moves |S11| at 915 MHz from -10.4 to -42.7 dB and keeps the whole US915 band below -21 dB.

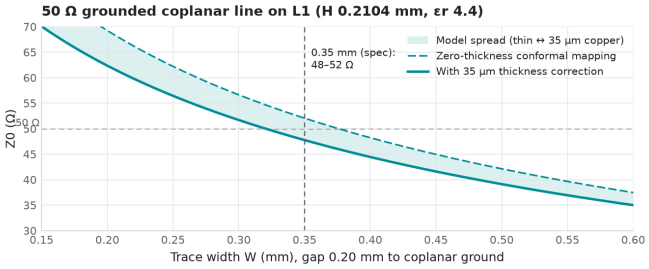


Figure 10. Grounded coplanar waveguide on L1 over the L2 ground (JLC 7628 stackup, 0.20 mm gap): 50 Ω at 0.32–0.376 mm. Rev B uses 0.35 mm (48–52 Ω).

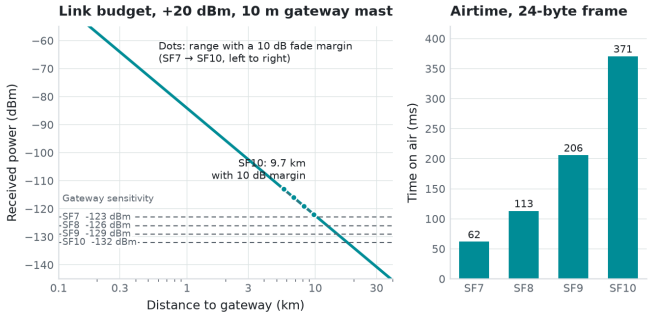


Figure 11. Log-distance path loss for a rural site, gateway at 10 m. With a 10 dB fade margin: SF7 5.7 km, SF9 8.1 km, SF10 9.7 km.

SPREADING FACTOR	SF7	SF8	SF9	SF10
Airtime, 24-byte frame	62 ms	113 ms	206 ms	371 ms
Range, 10 dB margin (model)	5.7 km	6.8 km	8.1 km	9.7 km
Charge per uplink at 3.3 V	9.0 mC	–	21.5 mC	35.9 mC

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4 · MAIN BOARD: TW1-MAIN REV B

Tradewind has **one custom circuit board**. Everything else electrical in the product is a bought sub-assembly on a cable: the panel, two probes, the rain gauge, the antenna and the cell. This section is the complete package for the board: schematic and design notes (4.1), electrical considerations (4.2), BOM (4.3), layout (4.4), EMI/EMC and ESD (4.5) and manufacturing outputs (4.6). Section 5 covers each bought sub-assembly: how it connects and what that means electrically.

ELECTRICAL ASSEMBLY	MADE OR BOUGHT	CONNECTS AT	COVERED IN
TW1-MAIN main board: charger, buck-boost, radio module, sensors, I/O	Designed here	–	Section 4
Solar panel, 5 V / 2 W, on the roof	Bought	J1 (JST PH 2)	5
SHT45 air probe, 1 m PUR cable, in the radiation shield	Bought	J4 (JST PH 4)	5
Capacitive soil probe	Bought	J3 (JST PH 3)	5
Tipping-bucket rain gauge (reed switch)	Bought	J2 (JST PH 2)	5
Antenna feed: U.FL pigtail, SMA bulkhead, 915 MHz whip	Bought	J5 (U.FL)	5
Protected 18650 cell (with its own protection PCB)	Bought	BT1 holder	5

Board at a glance

Size, build	96 × 70 × 1.6 mm, 4 layers, ENIG, matte black
Parts	85 footprints, 268 pads, all SMT on top except BT1
Nets	47 schematic nets; 3 net classes (Default, Power, RF)
Copper	497 track segments, 1,261 mm on L1, 174 mm on L4
Vias	119 (72 GND, 21 +3V3); 0.6/0.3 mm and 0.45/0.2 mm
Checks	ERC 0/0 · DRC 0 with schematic parity · netlist 233 pins, 0 mismatches

Net classes

CLASS	NETS	TRACK / CLEARANCE
Power	SOLAR_RAW, VIN_SOLAR, VBAT, VSYS, LX1, LX2	0.30 / 0.15 mm
RF	RF_MOD, RF_ANT	0.35 / 0.20 mm
Default	everything else	0.20 / 0.15 mm

+3V3 and GND are planes (L3, L2), plus pours on L1 and L4. The power class sets the minimum; the switch-node and plane connections are drawn wider by hand.

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4.1 Schematic

KiCad 9, hierarchical: a root sheet and three function sheets, 69 placed parts. ERC: 0 errors, 0 warnings. The netlist is checked pin by pin against the design source (233 pins, 47 nets, 0 mismatches) in `tools/check_netlist.py`.

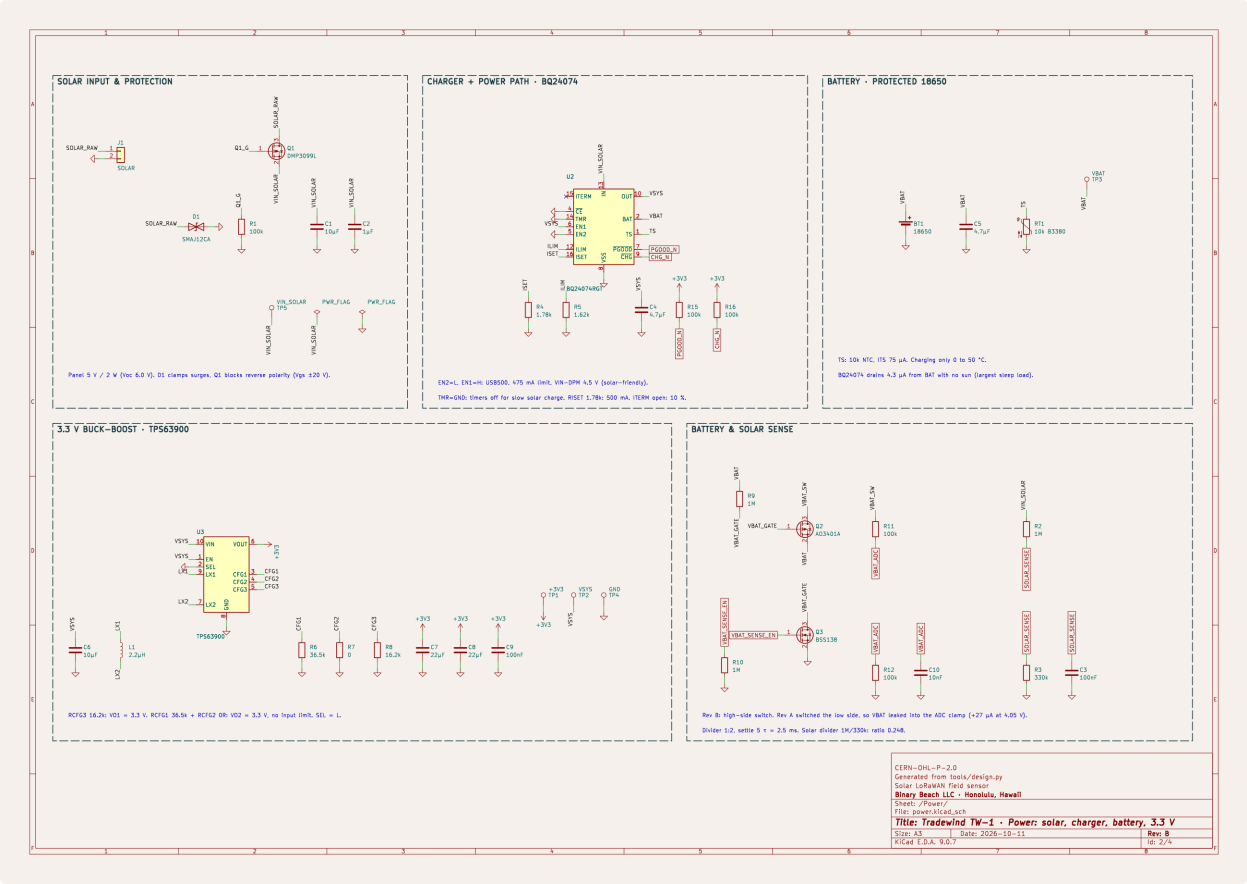


Figure 12. Sheet 2, Power: solar input and protection, BQ24074 charger and power path, protected 18650 with NTC, TPS63900 buck-boost, switched battery and solar sense.

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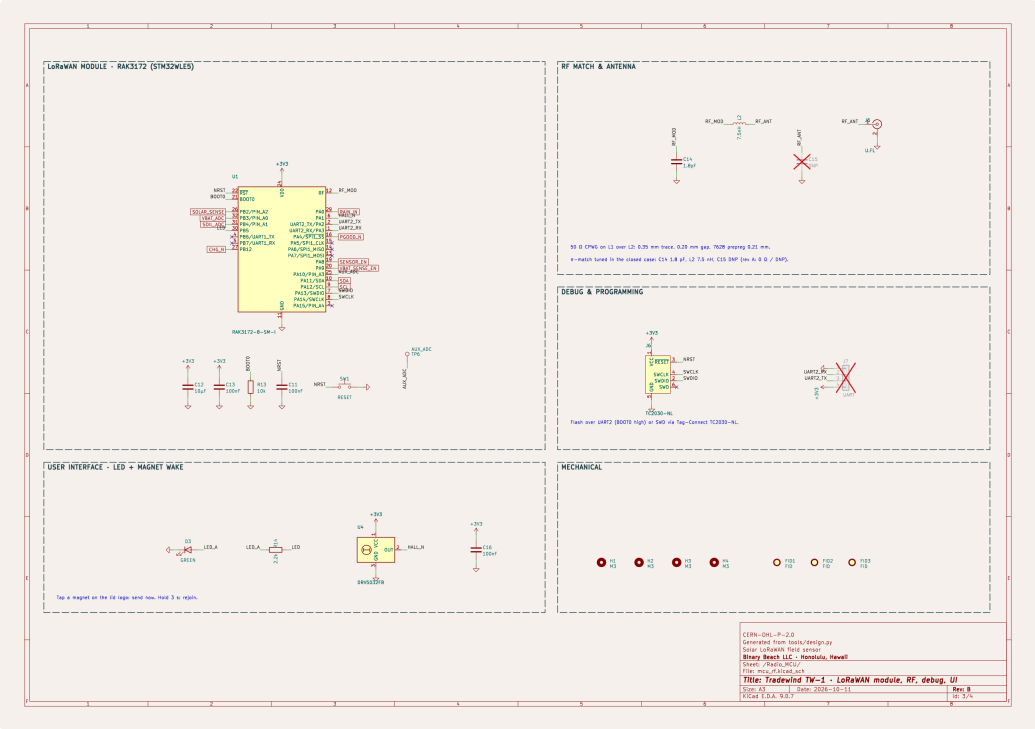


Figure 13. Sheet 3, Radio and MCU: RAK3172, π-match to U.FL, SWD (Tag-Connect) and UART pads, reset, LED and magnet wake.

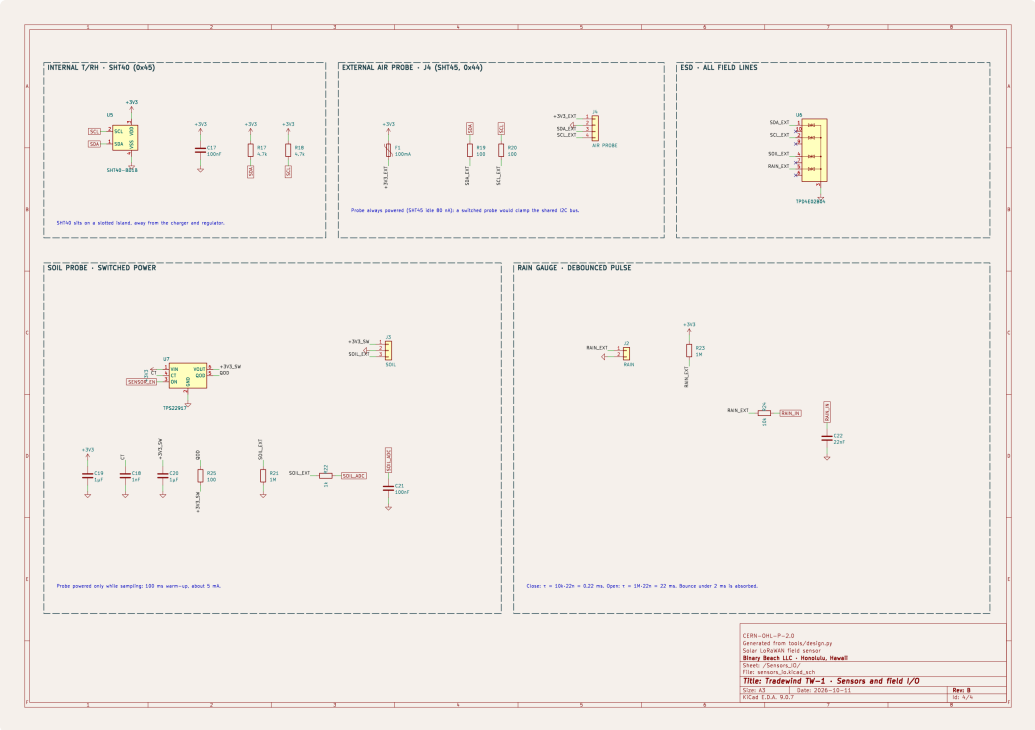


Figure 14. Sheet 4, Sensors and field I/O: internal SHT40, J4 air probe, J3 soil probe on a load switch, J2 rain gauge with debounce, and 4-channel ESD.

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Schematic notes, block by block

BLOCK	DECISION AND THE REASON FOR IT
Solar input J1, D1, Q1, C1, C2	D1 SMAJ12CA (bidirectional, 12 V standoff, 19.9 V clamp at 20 A) takes surges and ESD on the panel lead. Q1 DMP3099L is a P-FET in the high side with its gate pulled to GND (R1): a correct panel turns it on, a reversed panel leaves it off. It is rated −30 V with ±20 V on the gate, above both the clamp voltage and the panel's 6.0 V open circuit. A FET drops about 45 mV at the input limit; a Schottky would drop about 0.35 V, which the 4.5 V input-regulation point can't spare. C1 10 µF / 25 V keeps about half its value at 5 V of bias, inside the charger's 1–10 µF range for IN.
Charger U2 BQ24074	USB500 mode (EN1 = VSYS, EN2 = GND): the input is limited to 475 mA and **VIN-DPM** holds the panel at 4.5 V, which is the reason this part works without an MPPT chip. EN1 is tied to VSYS, not to the panel, because the EN pins are rated 7 V. ISET 1.78 kΩ sets 500 mA. TMR is grounded so the safety timers can't end a slow solar charge. The TS NTC limits charging to about 0–50 °C. ILIM is populated as the datasheet asks, though USB500 mode ignores it. CHG and PGOOD are open-drain with pull-ups to +3V3 (not VSYS), so the MCU pins never see more than VDD.
Charger heat	Worst-case linear-charger dissipation is (VIN − VBAT) × I ≈ (5.0 − 3.2 V) × 0.40 A ≈ 0.7 W, at a low cell with a strong panel. At 44.5 °C/W junction to air (datasheet, JEDEC board) that is a 31 °C rise, so about 91 °C in a 60 °C box. The charger's own thermal regulation (125 °C) cuts the current before it gets dangerous. U2 sits 76 mm from the internal humidity sensor, and the measured coupling is 3.1 °C per watt.
Buck-boost U3 TPS63900	The cell crosses 3.3 V during discharge, so the converter must both step up and step down. Its quiescent current is 75 nA. There is no feedback pin: the output is set by CFG resistors that are read once at power-up (R8 16.2 kΩ → 3.3 V). They are 1 %, 100 ppm parts on short traces, because a misread sets the wrong rail; the pins tolerate at most 10 pF and these traces add 0.2 pF. With the input-current limit set to unlimited, the converter's peak switch current is 1.33–1.6 A and TI asks for an inductor with Isat ≥ 2 A. L1 is a 2.2 µH molded metal-alloy part rated 2.6 A. C7 and C8 are 22 µF / 10 V X5R and keep about 60 % at 3.3 V, and the transmit-droop simulation (3.4) uses that derated value.
Battery sense Q2, Q3, R11, R12, C10	The rev A low-side switch leaked 27–40 µA through the ADC clamp. Rev B uses a high-side switch, which leaks 3 nA when off. After Q2 turns on, C10 charges through R11 ∥ R12 (50 kΩ), with τ = 0.5 ms. A 12-bit reading needs 4.5 ms to settle to ½ LSB. **Found in this review:** the firmware waited 3 ms, which reads 0.25 % low (≈ 10 mV at 4 V). It now waits 5 ms.
Radio / MCU U1 RAK3172	The module carries the FCC modular approval, the STM32WLE5 and its 32 MHz reference, all behind a shield can. VDD gets 10 µF + 100 nF at the pin. BOOT0 is pulled down, and NRST has 100 nF plus a reset switch inside the case. Debug uses a Tag-Connect footprint (no connector to buy) and UART pads for AT commands and the bootloader.
RF match C14, L2, C15	The π footprint carries a shunt C (C14 1.8 pF C0G ±0.1 pF), a series L (L2 7.5 nH, high-Q) and a spare shunt (C15, DNP). Values come from S11 measured in the closed case and the optimizer in `sim/rf_match.py`. Shunt-C plus series-L is a low-pass, so it also cuts the harmonics (4.5).
Air probe J4, F1, R19, R20	The probe stays powered (80 nA idle): an unpowered SHT45 would clamp the shared I²C bus through its ESD diodes. F1 (100 mA PTC) protects +3V3 from a crushed cable. R19 and R20 (100 Ω) limit ESD current into the MCU pins and damp ringing on the cable. The pull-ups (4.7 kΩ) sit on the board side.
Soil probe U7, R21, R22, C21	TPS22917 load switch: the probe is powered only for its 100 ms warm-up. CT 1 nF slows turn-on so the probe's input capacitance doesn't pull +3V3 down. R21 (1 MΩ) reads a missing probe as 0 V. R22 and C21 (1.6 kHz low-pass) remove the probe's own oscillator ripple.
Rain / magnet R23, R24, C22, U4	The 1 MΩ pull-up draws 3.3 µA only while the reed is closed. The 10 kΩ / 22 nF filter rejects bounce (τ_{fall} 0.22 ms, τ_{rise} 22 ms), and the 22 nF also bypasses RF picked up on the gauge lead. The DRV5032FB is push-pull, so it needs no pull-up current.

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4.2 Electrical design considerations

The board is small and slow by most standards; the hard limits are microamps in sleep, a 0.5 A linear charger, a +20 dBm transmitter and four cables that leave the box. Every number below is measured off the routed board by `tools/layout_audit.py` and evaluated in `sim/board_si.py`.

High-current paths

PATH	NET	WIDTH MM	LENGTH MM	R MΩ	I MAX MA	DROP MV	AT 60 °C	ΔT °C
Panel in → reverse FET	SOLAR_RAW	0.30	12.6	21	440	9	10	+1.6
Reverse FET → charger IN	VIN_SOLAR	0.30	32.6	53	475	25	29	+1.9
Charger BAT → cell +	VBAT	0.30	35.9	61	500	31	35	+2.1
Charger OUT → buck-boost VIN	VSYS	0.30	13.9	23	120	3	3	+0.1

1 oz outer copper; ΔT from the IPC-2221 external-layer curve. A 0.3 mm track carries 1.0 A for a 10 °C rise, so heating is never the issue; **resistance is**. The drops that matter: the 31 mV on VBAT during constant-current charge makes the charger enter constant-voltage about 31 mV early, which lengthens the charge slightly but costs no capacity (the drop falls to ~3 mV at the 50 mA termination current). VIN_SOLAR's 25 mV is harmless: VIN-DPM regulates at the IN pin, so the panel simply runs 25 mV higher on its curve. Rev C widens VBAT to 1 mm (9 mV).

Fast and high-frequency nets

NET	WHAT MAKES IT FAST	TREATMENT
RF_MOD / RF_ANT	915 MHz, +20 dBm	50 Ω CPWG, 14.1 mm, via fence, nothing else on L1 nearby
LX1 / LX2	Switch node, ns edges, pulsed amps	7.1 mm², L1 only, no vias, solid GND below
VSYS / PGND at U3	Hot-loop di/dt	Loops 1.1 and 3.4 mm²
32 MHz reference	Inside the module can	Nothing to route; it shows in the scan (4.5)
SWD, UART	≤ 4 MHz, bench only	Short, L1, near the module
SDA / SCL	100 kHz, but 1 m of cable	Rise time checked (below)

Sensitive nets

NET	WHY IT'S SENSITIVE	TREATMENT
CFG1–3	Read once; a misread sets VOUT	≤ 1.9 mm, 1 % / 100 ppm
ISET, ILIM	Set the charge current	Resistors at the pins
TS	NTC voltage sets the charge window	Kept 7+ mm from the switch node
VBAT_ADC	50 kΩ source	10 nF at the divider; trace driven from 10 nF
SOLAR_SENSE	248 kΩ source	100 nF at the divider
SOIL_ADC	Probe output, 1 m cable	1 kΩ + 100 nF at the MCU side

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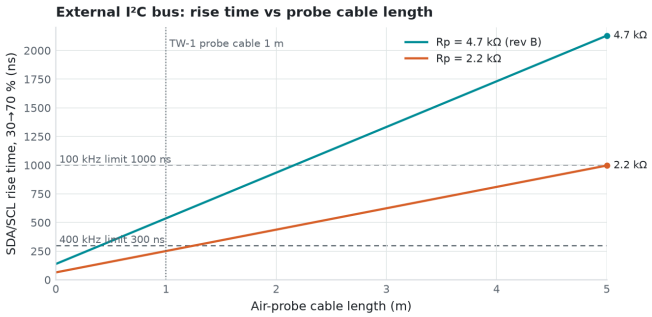


Figure 15. External I²C rise time. Bus capacitance with the 1 m probe ≈ 135 pF gives 536 ns with 4.7 kΩ: fine at 100 kHz (limit 1000 ns), too slow for 400 kHz. 100 kHz works to 2.1 m of cable.

I²C over the probe cable

The internal bus (SHT40, 87 mm of trace at 0.89 pF/cm) is 23 pF and rises in 91 ns. The 1 m probe cable adds about 100 pF per conductor. The firmware therefore clocks at **100 kHz**, which tolerates 2.1 m with the fitted 4.7 kΩ pull-ups. 400 kHz would need 2.2 kΩ and a cable under 1.2 m. That would gain nothing, because the bus moves 12 bytes per reading. SDA and SCL run side by side in the unshielded cable, so crosstalk is limited by keeping the edges slow: 4.7 kΩ pull-ups and 100 Ω series resistors.

Assumed: 5 pF/MCU pin, 10 pF/SHT4x, 0.25 pF ESD, cable 100 pF/m. Measure the cable before raising the clock.

ADC sources and settling

NET	SOURCE R	FILTER C	T	½ LSB AT 12 BIT	FIRMWARE WAIT	SAMPLE KICK-BACK
VBAT_ADC	50 kΩ	10 nF	0.50 ms	4.5 ms	5 ms	0.050 %
SOIL_ADC	1 kΩ	100 nF	0.10 ms	0.9 ms	100 ms	0.005 %
SOLAR_SENSE	248 kΩ	100 nF	24.81 ms	223.6 ms	always on	0.005 %

Each divider has a filter capacitor large enough that the ADC's ~5 pF sampling capacitor takes at most 0.05 % of its charge, so the 50–250 kΩ dividers need no buffer amplifier. SOLAR_SENSE is never switched; its divider draws 4.5 µA from the panel, never from the cell.

Plane pair and decoupling

L2 (GND) and L3 (+3V3) are 1.07 mm apart, which gives only 242 pF of plane capacitance. The design doesn't rely on it. Decoupling is by capacitors at each supply pin: 10 µF + 100 nF at the module, 100 nF at each sensor and the hall switch, and 22 µF × 2 at the converter. The module's TX step (5 → 87 mA in ~10 µs) is carried by C12 and the converter output. Simulated droop is 32.6 mV; measured 40 mV.

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4.3 Bill of materials and cost

47 line items, 69 placements, 46 unique parts; all single-side SMT except the battery holder. Full list with manufacturers, footprints and LCSC numbers: `hardware/fab/BOM_tradewind_revB.csv` (also in JLCPCB format).

REFS	PART	MPN	QTY	@100	@1K
U1	LoRaWAN module, STM32WLE5CC, US915, RF pad (IPEX via board	RAK3172-8-SM-I	1	$6.90	$5.950
BT1	18650 holder; protected Li-ion 3350 mAh cell	1042	1	$2.45	$1.900
U2	1-cell Li-ion linear charger with power path, VIN-DPM	BQ24074RGTR	1	$1.55	$1.100
U3	Buck-boost 1.8–5.5 V in, 75 nA Iq	TPS63900DSKR	1	$1.45	$1.050
U5	T/RH sensor, I2C 0x45, inside the case (condensation / lea	SHT40-BD1B-R2	1	$1.35	$0.980
SW1	Tactile switch, reset (inside the case)	B3U-1000P	1	$0.50	$0.380
U4	Hall switch, omnipolar, 5 Hz, 0.54 µA (magnet tap)	DRV5032FBDBZR	1	$0.45	$0.320
J5	U.FL receptacle to SMA bulkhead pigtail	U.FL-R-SMT-1(10)	1	$0.33	$0.220
U6	4-ch ESD, ±12 kV contact, 0.25 pF	TPD4E02B04DQAR	1	$0.31	$0.210
U7	Load switch, 2 A, 10 nA leakage (soil probe power)	TPS22917DBVR	1	$0.29	$0.190
J4	External SHT45 probe (radiation shield)	S4B-PH-SM4-TB(LF)(SN)	1	$0.25	$0.180
J3	Capacitive soil-moisture probe	S3B-PH-SM4-TB(LF)(SN)	1	$0.23	$0.170
… plus 35 lines of passives and connectors					

COST PER UNIT	10 UNITS	100 UNITS	1,000 UNITS
Components	$23.71	$17.98	$13.83
+ bare board (4-layer, ENIG) + SMT assembly	$44.21	$23.28	$16.43
+ protected 18650 cell	$51.71	$29.48	$21.23
+ enclosure, panel, shield, bracket, glands, antenna	**–**	**$90.44**	**$54.86**

Prices are typical distributor quantity breaks (October 2026). The 1k mechanical cost assumes molded box and lid; tooling (about $7–9k) is not included. The radio module is 43 % of the parts cost at 1k; a chip-down STM32WL design would save about $3.50 per unit but needs its own FCC certification.

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BOM considerations

PART	SOURCING	SUBSTITUTION RULE
U1 RAK3172-8-SM-I	Single source	**None.** The FCC modular grant and the RF match belong to this module. A different module means a new grant review and re-tuning.
U2 BQ24074RGT	Single source (TI)	Not even within the family: the bq2407x variants differ in OVP and VO(REG), and this design depends on the 10.5 V OVP and VIN-DPM in USB500 mode.
U3 TPS63900DSK	Single source (TI)	None at this quiescent current. The fallback is a buck-boost with a feedback divider, which costs about 1 µA of sleep current.
U5 SHT40-BD1B, probe SHT45	Sensirion	The "BD1B" suffix sets I²C address 0x45, so the internal sensor and the probe (0x44) don't collide. Order by full part number.
C14, L2	Murata	**Exact series only:** C0G ±0.1 pF and LQG15HS high-Q. Generic ±0.25 pF or low-Q parts move the match by several dB.
L1 2.2 µH 2520	Multi-source	Same size, Isat ≥ 1.5 A, DCR ≤ 100 mΩ, shielded or molded construction.
R6–R8 (CFG), R4 (ISET), RT1	Multi-source	1 %, ≤ 100 ppm/°C. RT1 must be B ≈ 3380–3435 or the charge window moves.
Q1, Q2, Q3, D1, passives	Multi-source	Ratings in the derating table below; 0402 / 0603 / 0805 footprints are generic.

Derating and temperature

PART	RATED	WORST CASE IN USE	RATIO
C1, C2 on VIN_SOLAR	25 V	19.9 V TVS clamp (6.0 V Voc)	80 % (24 %)
Q1 V_{DS} / V_{GS}	−30 V / ±20 V	19.9 V clamp / 6.0 V	66 % / 30 %
D1 standoff	12 V	6.0 V open circuit, cold	≈ 55 %
C4–C8 (10 V)	10 V	4.4 V (VSYS)	44 %
L1 saturation	2.6 A	1.6 A peak-current limit, max	62 %
U3 VIN (abs max)	5.9 V	4.4 V VSYS (VO(REG))	75 %
U2 BAT (abs max)	5 V	4.2 V	84 %
F1 hold	100 mA	probe ≤ 1 mA measuring	1 %
VBAT track	1.0 A @ +10 °C	0.5 A	50 %

Every IC and passive is rated to at least +85 °C (X5R and X7R dielectrics). The box interior peaked at 37.1 °C in the field; the chamber limit is +60 °C. MSL: the module, U2 and U3 are moisture-sensitive. Follow the reel-label floor life and bake before rework. The cell is shipped at 30 % state of charge (UN 38.3; ships as UN 3481 packed with equipment).

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4.4 PCB layout

LAYER	USE	THICKNESS
L1 F.Cu	Signals, RF CPWG, power stage, GND pour	35 µm
Prepreg 7628	εr 4.4	0.21 mm
L2 In1.Cu	**Solid GND**: no splits, no traces	15 µm
Core	FR-4	1.07 mm
L3 In2.Cu	+3V3 plane	15 µm
Prepreg 7628	εr 4.4	0.21 mm
L4 B.Cu	Slow signals + GND pour	35 µm



Rev B top side, KiCad raytrace.

JLC 4-layer standard: 0.15 mm track and space, 0.2 mm drill, 0.3 mm to edge. DRC with schematic parity: **0 errors**. L1 sits 0.21 mm over the ground plane, so every top-layer trace has a tight, unbroken return path directly beneath it. That is the main reason for choosing 4 layers on a board this slow.

TW-1 rev B copper by electrical role (top layer solid, bottom layer faint)

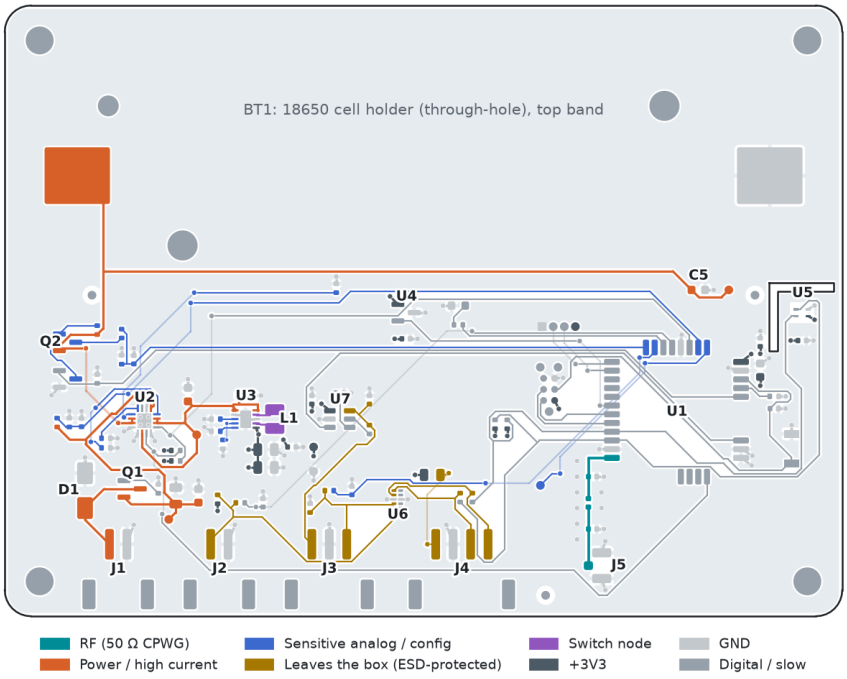


Figure 16. The routed board colored by electrical role (from the board file). Power runs bottom-left, field I/O bottom-middle, RF bottom-right; the cell occupies the top band and VBAT crosses under it to C5.

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Placement and zoning

- **One cable wall.** J1–J5 line the bottom edge, so every cable leaves through the box's bottom wall (four glands and the SMA bulkhead). Water runs off the glands, and cable currents stay along one edge, far from the RF line.
- **Noisy and quiet zones.** Power (charger and buck-boost) sits bottom-left beside J1, field I/O bottom-middle, and RF bottom-right next to J5. The switch node is 36 mm from the RF line, 40 mm from the U.FL and 62 mm from the internal humidity sensor.
- **Thermal.** The two heat sources (charger U2, module U1 during transmit) are 76 mm and 18 mm from U5. U5 also sits on a peninsula cut by an L-shaped milled slot, with no copper pour on any layer under it.
- **Cell band.** The 18650 holder takes the top 33 mm. The NTC (RT1) sits against the cell body, so the charger sees the cell's temperature, not the board's.

Layer usage and return paths

- L2 is unbroken GND under the whole board; only via barrels perforate it. Every L1 trace returns directly beneath itself.
- L4 traces reference the **+3V3** plane (L3), not ground. That return path closes through decoupling capacitors, so only slow nets are allowed on L4. The 14 nets with copper there are DC or kHz signals: +3V3_EXT, AUX_ADC, CFG2, CHG_N, HALL_N, ILIM, PGOOD_N, RAIN_IN, SOIL_ADC, SOLAR_SENSE, TS, UART2_RX, UART2_TX, VBAT. RF, the switch node and I²C stay on L1.
- Plane connections: every GND and +3V3 pad has its own via (72 GND, 21 +3V3), so no part shares a via with a neighbour's return current. U2's pad has 4 thermal vias and U3's has 6, both to L2.

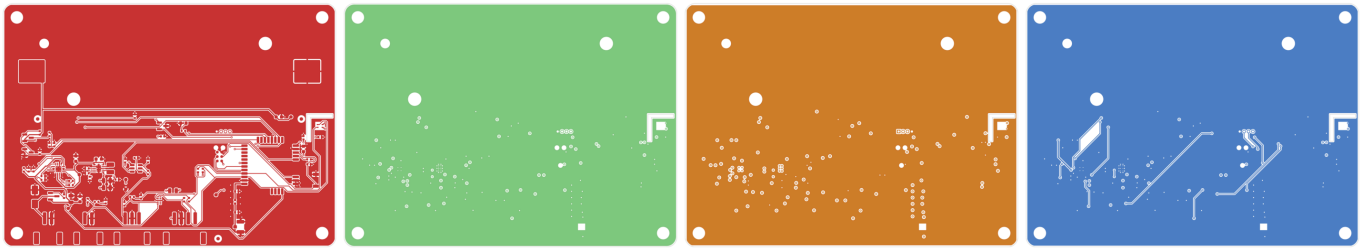


Figure 17. Copper layers L1–L4 (left to right): top signals and GND pour, solid ground, +3.3 V plane, bottom signals and GND pour.

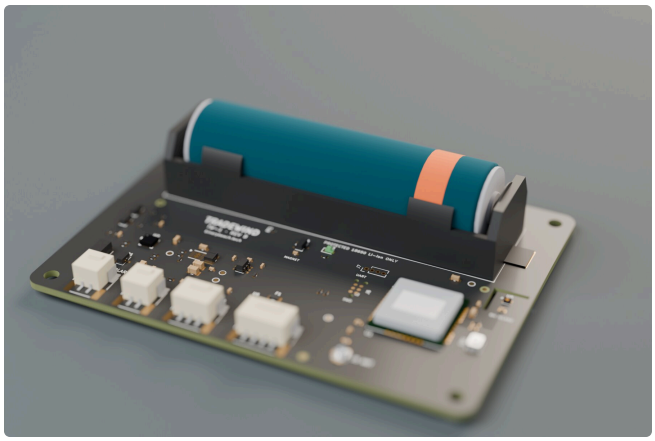


Figure 18. Populated rev B board (Blender Cycles render from the KiCad 3D export).



Figure 19. RF corner: RAK3172, π-match on the CPWG line, via fence and U.FL.

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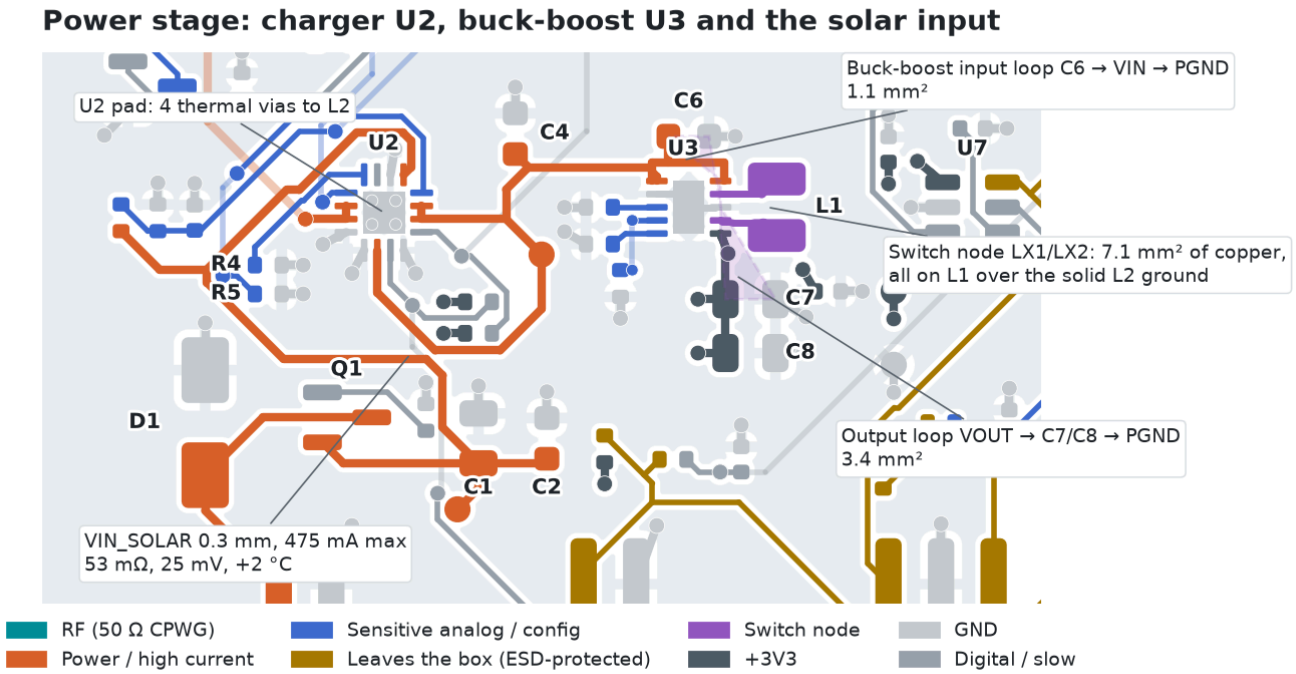


Figure 20. Power stage as routed: charger U2 with its thermal vias, buck-boost U3 with the two hot loops marked, and the solar input path from Q1.

Buck-boost (hand-routed)

- Placed and routed per TI's layout guidance before autorouting, then locked.
- Input loop C6 → VIN → PGND: **1.1 mm²**. Output loop VOUT → C7 → PGND: **3.4 mm²**. A buck-boost runs both loops hot (buck below 3.3 V, boost above), so both are kept small.
- Switch node: 7.1 mm² of copper, enough to carry the current and no more, all on L1 over solid ground, with no vias.
- CFG1–3 are under 2 mm long. CFG2's one hop to L4 sits away from LX.

Charger and input

- U2 has 4 thermal vias under its pad into the L2 plane, which spreads the heat across the board.
- C4 (OUT) is 6.2 mm of track from U2, and C6 is 3.8 mm from U3.
- VIN_SOLAR is 0.3 mm and 33 mm from Q1 to U2, a detour left by the autorouter. It costs 25 mV, which VIN-DPM absorbs.
- **Found in this review:** TI wants the IN and OUT/BAT capacitors close to the IC. C4 is; C2 and C5 are not. Rev C fixes both; see the findings table.

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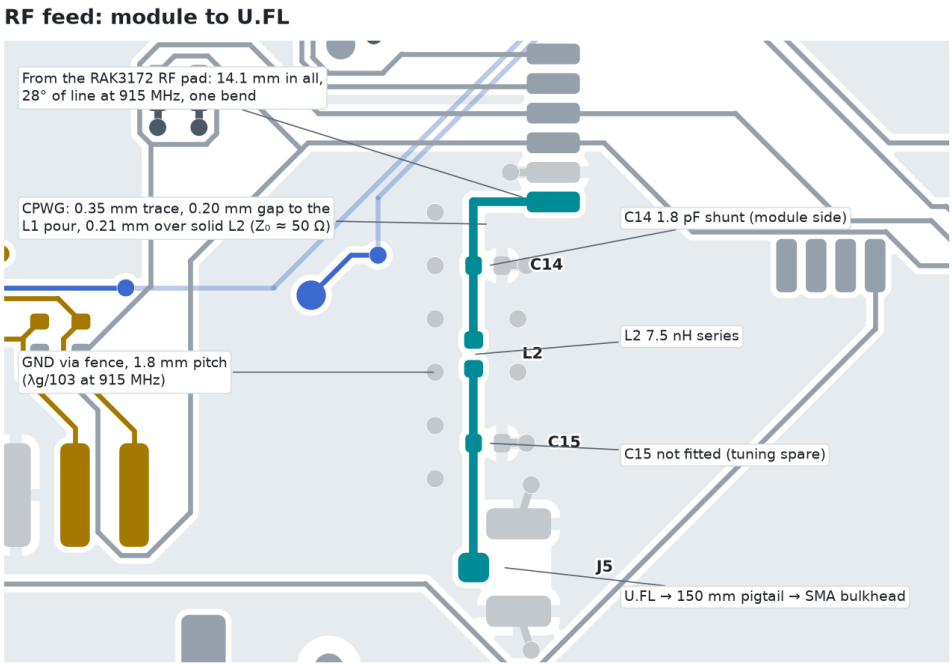


Figure 21. RF feed as routed: the module RF pad, the π-match pads on the line, the GND via fence and the U.FL at the board edge.

RF layout

- CPWG on L1: 0.35 mm trace with a 0.20 mm gap to the pour, over solid L2 0.21 mm below. Field-solver range 47.8–52.1 Ω (3.5).
- 14.1 mm from the module pad to the U.FL, 28° at 915 MHz, one bend. The match pads sit directly on the line, with no stubs.
- 11 GND vias line it (6 on one side, 5 on the other, including the C14/C15 ground vias), at a 1.8 mm pitch (λg/103). That is far tighter than the λ/20 rule even at the 3rd harmonic.
- No other L1 copper runs near the line. One slow L4 trace crosses under the area, shielded by L2.
- J5 sits at the board edge in line with the SMA bulkhead, so the 150 mm pigtail runs straight with no tight bends.

Analog, sensing and debug

- Divider filter capacitors sit at the dividers, so the long ADC traces (VBAT_ADC 65 mm) are driven from a 10–100 nF source and pick up almost nothing.
- The ESD array U6 sits between J2–J4 and everything else. Its ground pins go straight to vias, so a strike's current returns to L2 before it can reach the MCU.
- SWD uses a Tag-Connect footprint next to the module; UART uses 1.27 mm pads.
- Test points: TP1 +3V3, TP2 VSYS, TP3 VBAT, TP4 GND, TP5 VIN_SOLAR, TP6 AUX_ADC. These are the bed-of-nails points for the production tester (section 8).
- Three fiducials; 0402 is the smallest part; the battery holder is the one through-hole part (hand or selective solder).

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Layout review findings (rev B as built)

Found by measuring the routed board (`tools/layout_audit.py`) after the rev B build. None of them shows up in the test data. All five are cheap to fix, and they are part of rev C.

FINDING	EFFECT	REV C ACTION
C5 (charger BAT capacitor) is 96 mm of track from U2 (160 mΩ, ≈ 96 nH)	The cell masks it in normal use. With no cell, or with the cell's protection open, the charger's BAT pin has no local capacitor.	Move C5 within 2 mm of U2 pins 2–3
C2 (charger IN capacitor) is 29 mm of track from U2	Hot-plugging the panel rings longer at the IN pin. That stays well inside the 26 V rating, because D1 clamps.	Move C2 next to U2 pin 13; leave C1 at Q1
VBAT is 0.3 mm wide, 36 mm long (61 mΩ)	31 mV drop at 500 mA charge; it enters constant-voltage early	1 mm track or a pour (9 mV)
C6 (U3 input) is 10 µF / 10 V 0603: about 4–5 µF left at 4.4 V of bias	At the 5 µF effective minimum in the TPS63900 datasheet. Bring-up ripple and droop were in spec.	22 µF / 10 V 0805
Battery-sense settle time 3 ms in firmware (τ = 0.5 ms)	VBAT reads 0.25 % low	Done: firmware waits 5 ms

4.6 Manufacturing outputs

FILE	CONTENTS
tradewind_revB_gerbers.zip	11 Gerber X2 layers, Excellon PTH/NPTH, drill maps, job file
tradewind_revB_cpl.csv	Pick-and-place, top side, DNP excluded
BOM_tradewind_revB.csv / BOM_JLCPCB.csv	Costed engineering BOM; assembler format with LCSC numbers
assembly_drawing_top.pdf	Fab-layer assembly drawing with references
tradewind_revB.step / .glb / .wrl	3D board with all parts (enclosure fit, web viewer)
erc_report.txt / drc_report.txt	Clean ERC and DRC (with schematic parity), generated on every export
layout_audit.json	Per-net copper, current paths, loop areas, separations and RF geometry, measured from the board file

The whole chain runs from scripts: `tools/design.py` → schematic, board, routing, DRC, Gerbers, BOM, then the layout audit. A design change is one edit and one command, and the schematic and board can't drift apart.

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4.5 EMI, EMC and ESD

The enclosure is unshielded ASA, so the board and its cables have to be quiet on their own. The design was laid out by naming each noise source, each victim and the path between them, and then cutting the path.

SOURCE	VICTIM	PATH	WHAT BREAKS THE PATH
Buck-boost switch node and hot loops	Radio receiver; cables as antennas	Near-field H-field; edges conducted onto VSYS and +3V3	Loops ≤ 3.4 mm², molded inductor, solid L2, 36 mm to RF; converter in pulse mode at light load
Radio TX, +20 dBm, and its harmonics	Restricted bands ($3f_0$–$5f_0$); sensor leads	Antenna; pickup on cables near the whip	π-match low-pass (Figure below); whip points down, away from the leads; 22 nF / 100 nF at the input pins
32 MHz reference and MCU clocks	FCC 15.109 limits	Module can seams; ground currents into cables	Module can; solid L2 under the module; probe cable 100 Ω series
Soil-probe oscillator	Radio receiver; SOIL_ADC	Probe cable	Powered only 100 ms per uplink, never during TX or RX; 1 kΩ + 100 nF filter
ESD and surge on the cables	MCU pins, charger IN	Conducted along the leads	U6 at the connectors, 100 Ω on I²C, D1 on the panel, PTC on the probe supply

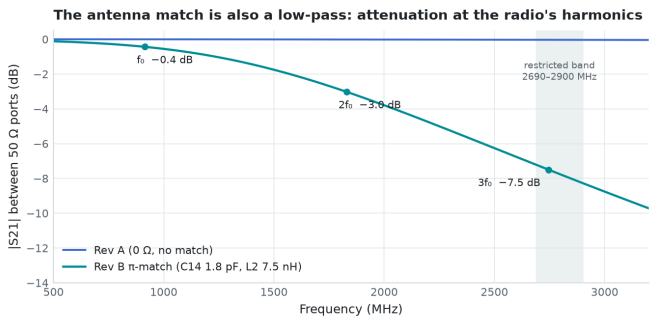


Figure 22. Simulated: C14 + L2 between 50 Ω ports cut $2f_0$ by 3.0 dB and $3f_0$ by 7.5 dB relative to rev A's 0 Ω. The 3rd harmonic of 902–915 MHz lands in the 2690–2900 MHz restricted band.

Regulatory path

The RAK3172's FCC modular grant covers the transmitter, as long as the antenna is of the type and gain listed in the grant. The host still needs two things before sale: (1) Part 15.109 class B verification of the whole product as an unintentional radiator (SDoC), and (2) a check that the radio's spurious emissions, measured in this host, still meet 15.247 / 15.209 (KDB 996369). Pre-scans of both are below; the accredited runs are booked with the IP66 test.

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Pre-compliance results

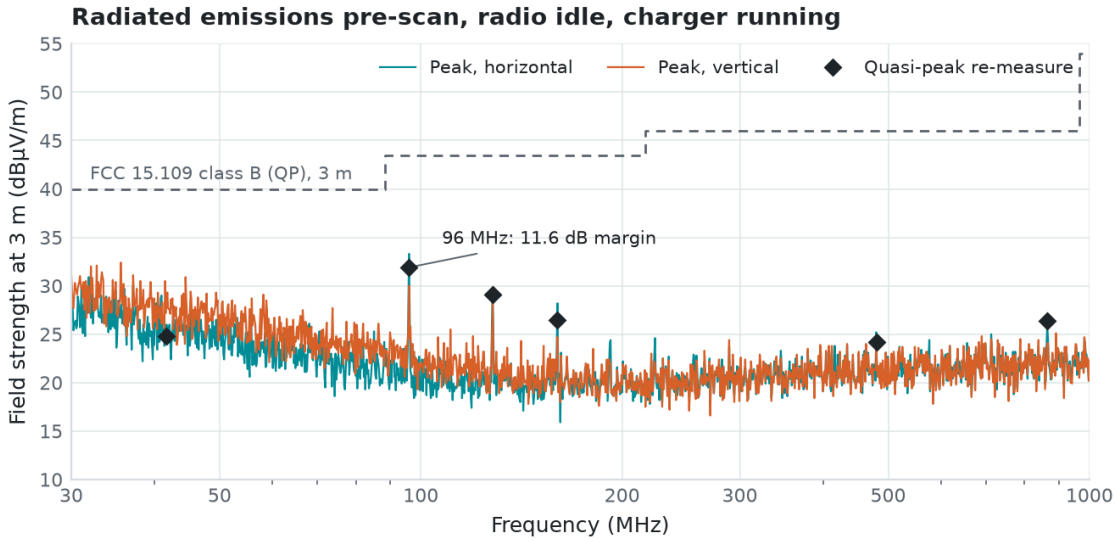


Figure 23. Radiated emissions, 3 m, charger running and radio idle. The worst quasi-peak is 96 MHz (RAK3172 32 MHz reference, 3rd harmonic), 11.6 dB under the class B limit. The 30–60 MHz hump is the converter's switch-node ringing coupling out on the probe cable.

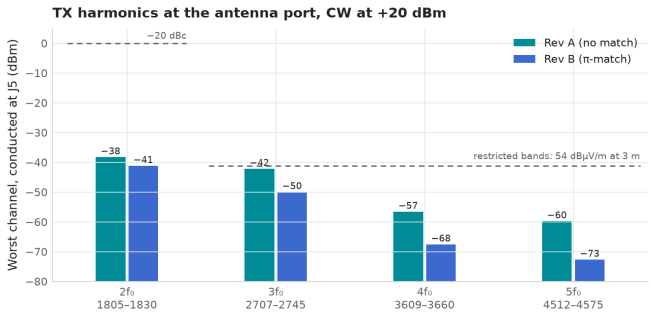


Figure 24. TX harmonics at J5, worst of 3 channels. Rev A's 3rd harmonic had 1.0 dB of margin to the restricted-band limit (0 dBi assumed at 2.7 GHz). Rev B's match lowers it 7.8 dB (simulated 7.5 dB), for 8.8 dB of margin.

RECEIVER DESENSE, SF10 BW125	WORST OF 3 CH
Reference: bench LDO into TP1, U3 off	-132.1 dBm
TPS63900 from the cell	+0.5 dB
+ charger at 400 mA	+0.4 dB
+ soil probe powered	+0.6 dB

ESD, IEC 61000-4-2	RESULT
±8 kV contact, J1–J4 and SMA shell	criterion A
±15 kV air, all ports and seams	A; B on J4

The board's own noise costs the receiver at most 0.6 dB, which is a few hundred metres of range at SF10. The one ESD reset (±15 kV air on J4) recovered by itself in 9 s.

Pre-compliance runs at a rented chamber and on the bench, not accredited tests; method and setup are in the log headers in `test/data/`.

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5 · OFF-BOARD ELECTRONICS: CABLES AND BOUGHT SUB-ASSEMBLIES

These parts are bought complete, so there is no board of ours to lay out in them. What matters electrically is how each one meets the main board: the cable, the protection at the connector, and what it costs in sleep current.

SUB-ASSEMBLY	INTERFACE	ELECTRICAL CONSIDERATIONS	PROTECTION ON TW1-MAIN
Solar panel 5 V / 2 W	J1, 2 wires, ≤ 0.44 A, Voc 6.0 V	Hot-plugged in the field, so the lead rings at connection. Reversal is possible on a field repair. The epoxy panel has no metal frame to ground.	D1 TVS, Q1 reverse-polarity FET, charger IN takes 26 V without damage
SHT45 air probe	J4, 4 wires, I²C at 100 kHz, 1 m unshielded PUR	The longest digital run and the main antenna for noise. Always powered (80 nA) so the bus can't be clamped. Rise time 536 ns at 1 m. It took the one ESD soft reset (±15 kV air).	U6 ESD, R19/R20 100 Ω, F1 PTC on +3V3_EXT
Soil probe (capacitive)	J3, 3 wires, analog 0–3 V, ~5 mA while on	Contains its own oscillator, which is noisy on a cable, so it is powered only 100 ms per reading and never while the radio is active. The output is high-impedance; a missing probe reads 0 V.	U7 load switch with soft start, U6, R22/C21 filter, R21
Rain gauge (reed)	J2, 2 wires, contact closure, up to a few metres	Bounce gave 2–6 counts per tip in rev A. The long lead can pick up RF and surges. A gauge parked with the reed closed draws 3.3 µA.	U6, R24/C22 filter (also an RF bypass), firmware lockout
Antenna feed	J5 U.FL → 150 mm 1.13 mm coax → SMA bulkhead → whip	Pigtail loss 0.31 dB, included in the TX-power numbers. The bulkhead shell is the one metal part through the box wall, so it takes ESD directly to the board's ground. The case detunes the whip, which the π-match corrects.	π-match, GND fence; ±8 kV on the SMA shell passed
Protected 18650	BT1 holder, 3.0–4.2 V, ≤ 0.5 A	The cell's own protection board opens at 2.5 V and on over-current. **A reversed cell is not blocked electrically:** a series FET would cost 20–40 mV and sleep current. It is controlled by the holder marking, the assembly step and production test.	RT1 NTC on the cell, TS charge window, silk polarity (rev C)

If the probe supplier changes, a custom probe-head board (SHT45 + ESD + connector on a 10 mm round PCB) would be the second board in the project. It is not needed for rev C.

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6 · FIRMWARE

RAKwireless RUI3 (Arduino-style API) on the module's STM32WLE5. LoRaWAN 1.0.3 OTAA, US915 sub-band 2, ADR on. The payload packer and the network-server decoder share golden test vectors: 16 C++ and 22 JavaScript host tests pass.

Per-uplink sequence

1. RTC alarm wakes from STOP2 (7.2 µA board total).
2. SENSOR_EN on, then wait 100 ms for the soil probe. Read both SHT4x (CRC-checked).
3. VBAT switch on, wait 5 ms, read the ADC. Also read panel voltage, CHG and PGOOD.
4. Pack 11 bytes, then send unconfirmed on fPort 1 at SF9 (206 ms at +20 dBm).
5. RX1/RX2 windows, then apply any downlink (interval, rapid mode, health, reboot).
6. Sleep until the next deadline. Rain tips and magnet taps wake it early.

Uplink payload, fPort 1 (11 bytes)

BYTE	FIELD	ENCODING
0	Flags	rapid · charging · PGOOD · low batt · sensor OK ×3
1–2	Air temperature	s16, 0.01 °C
3	Air humidity	u8, 0.5 %RH
4	Soil moisture	u8, 0.5 %
5–6	Rain tips	u16 cumulative, 0.2 mm/tip
7	Battery	2500 mV + 10 mV × n
8–9	Internal T / RH	s8 0.5 °C · u8 0.5 %RH
10	Panel voltage	u8, 25 mV

fPort 2: daily confirmed health frame (uptime, counters, RSSI/SNR, min battery, max internal temperature, charge minutes).

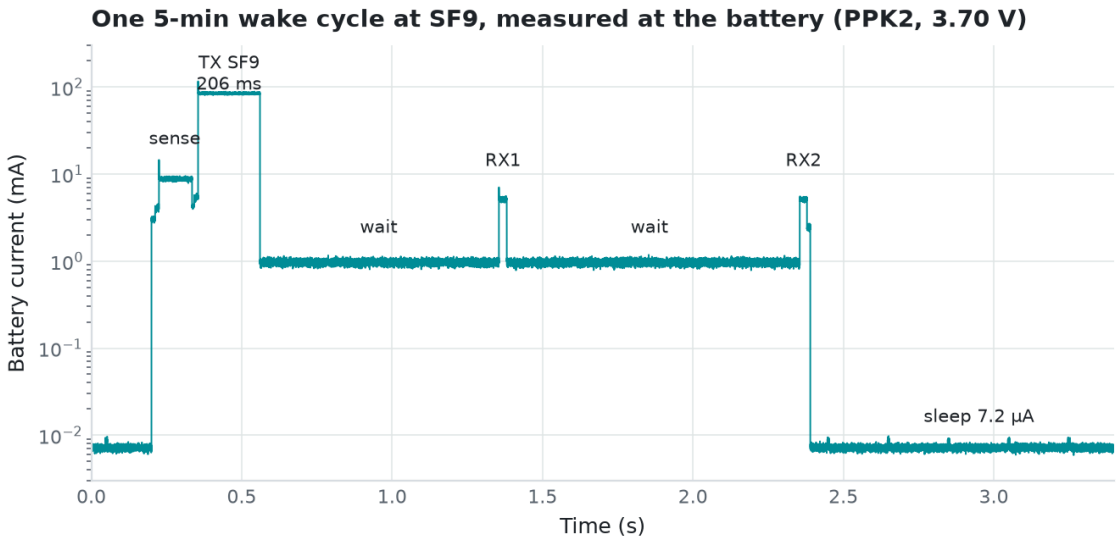


Figure 25. Measured current through one 5-minute wake cycle (Nordic PPK2, board TW1B-003). 20.6 mC per uplink against 21.3 mC predicted; awake 2.19 s.

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7 · ENCLOSURE AND MECHANICAL DESIGN



Figure 26. Exploded assembly: panel, ventilated roof and gussets, lid with light pipe, gasket, PCBA, base with heat-set inserts, glands, bracket and radiation shield.

ITEM	DESIGN
Box	122 × 98 × 45 mm, white ASA (UV-stable), 3 mm walls, 1° draft, ribbed lid
Seal	2.5 mm silicone cord in a 3.4 × 2.6 mm groove: 26 % squeeze, 78 % fill
Fasteners	4 × M4 captive lid screws into brass inserts; 4 × M3 PCB screws
Penetrations	4 × PG7 glands at 21 mm pitch, SMA bulkhead, M12 ePTFE vent. All on the bottom wall or side, none on top.
Sun roof	Carries the 136 × 110 mm panel, tilted 20°, 25 mm air gap over the box
Radiation shield	6-plate louvred stack, SHT45 probe holder, on the bracket arm
Mass	292 g box unit; 952 g installed
Envelope	285 × 362 × 149 mm installed
Prototype	FDM in ASA, 8.8 h for a full set

Fit checked in CAD against the board STEP: board-to-wall 2.4 mm, cell-to-lid 5.8 mm, gland nuts 10.2 mm. No interference in 12 checked pairs.

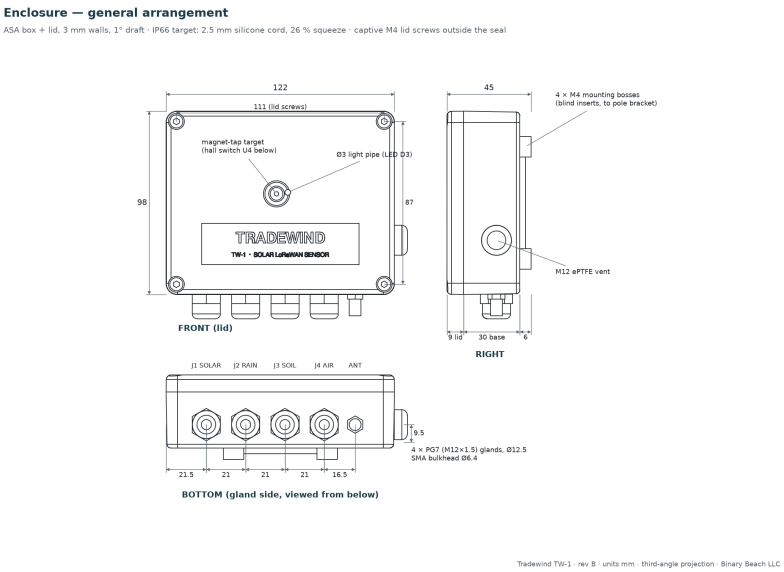


Figure 27. General arrangement of the enclosure: front, side and bottom views with overall dimensions and gland spacing.

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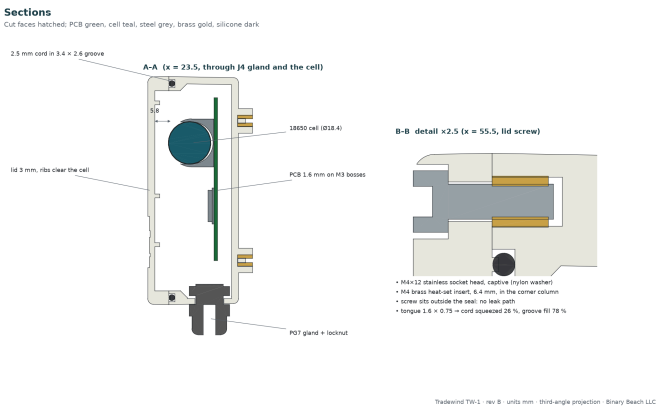


Figure 28. Sections A–A and B–B: PCB on bosses, cell under the lid ribs, gasket tongue and groove, lid screw and insert.



Figure 29. Lid off: PCBA on four bosses, gasket, glands and the U.FL to SMA pigtail.

Design for the field

- **Water.** All cables leave downward through glands, with drip loops; the antenna points down; the vent sits on a side wall below the gasket line.
- **Heat.** White ASA (solar absorptance ≈ 0.25) under a roof that blocks direct sun on the top and sides. Measured in the field: interior 3.8 °C above air on average, 5.4 °C worst.
- **UV.** ASA coupons changed ΔE 1.7 after 500 h of UV-A, against 8.9 for PETG, which is why PETG was rejected.
- **Production.** The walls are drafted and evenly thick so the printed shapes carry straight into injection-molding tooling. The roof and shield plates are flat-pack.

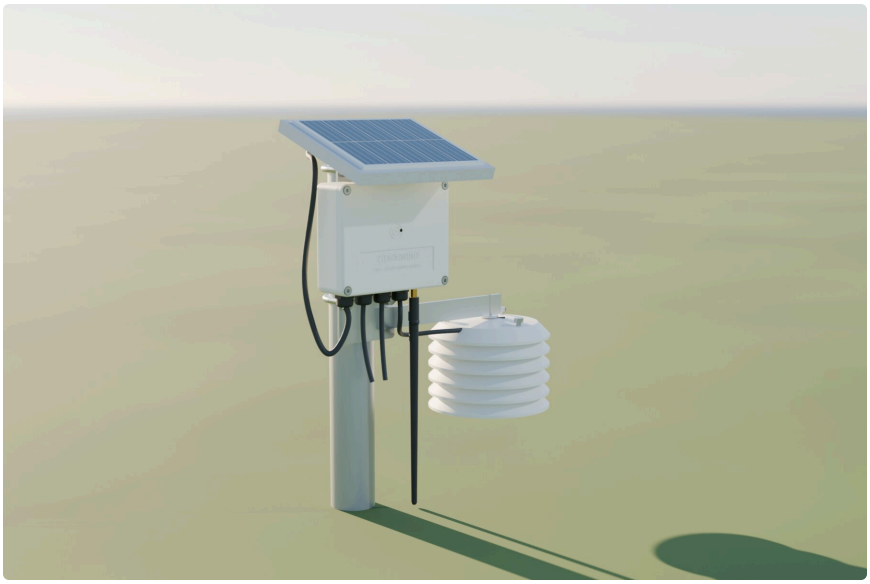


Figure 30. Installed on a 42 mm pole in daylight (render).

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8 · ASSEMBLY AND BRING-UP

Rev A: 3 boards, hand-assembled (stencil, hot plate, rework station). Rev B: 10 boards assembled by the board house. The full step-by-step log is in `test/BRINGUP.md`.

REV A ISSUE	HOW WE FOUND IT	ROOT CAUSE	REV B FIX (VERIFIED)
Sleep 34 µA (budget 7)	PPK2, part-lifting; current scaled with VBAT	Low-side divider switch: PB3 sat at VBAT and leaked through the clamp diode	High-side P-FET switch (Q2/Q3). 7.2 µA mean on 10 boards
2–6 rain counts per tip	50 hand tips counted 173	Reed bounce on an unfiltered EXTI input	10 kΩ / 22 nF filter + 50 ms firmware lockout. 1199/1200 on the bench
S11 −11 dB at 915 in the case	NanoVNA through the U.FL with the lid closed	Plastic and pigtail detune the whip	π-match C14 1.8 pF / L2 7.5 nH: -32.7 dB

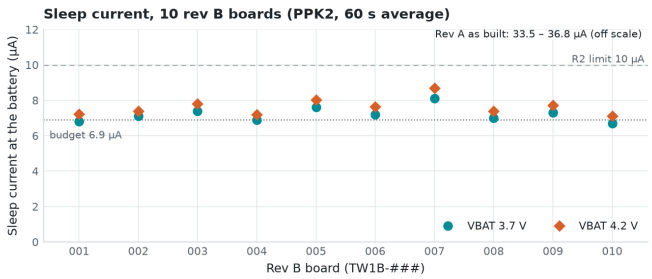


Figure 31. Sleep current, 10 rev B boards at 3.7 V: mean 7.21 µA, max 8.1 µA, against the rev A as-built values.

BRING-UP CHECK (10 BOARDS)	RESULT
+3V3	3.290–3.315 V
+3V3 ripple at 10 mA	20.7 mVpp max
VSYS in sun	4.394–4.438 V
Charge termination	4.194–4.213 V
Flash over UART2	11.7 s
OTAA join, median	6.0 s
Production test, first pass	90 % → 100 %

The one first-pass failure: TW1B-006, a solder bridge on U5 pins 3–4 (I^2C 0x45 missing). Reworked; rev C opens the paste aperture on that footprint.

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9 · CHARACTERIZATION

9.1 Power

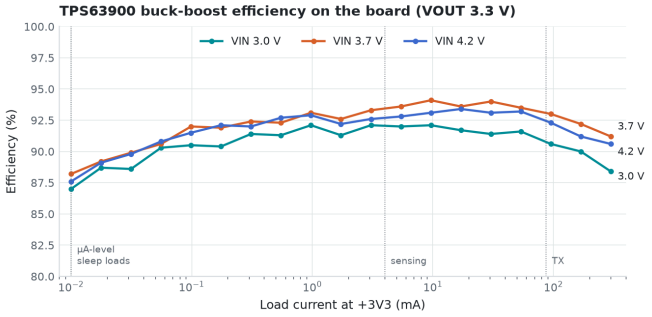


Figure 32. TPS63900 efficiency on the board: 88.2 % at 10 µA, 94.1 % at 10 mA, 92.9 % at 100 mA (3.7 V in). The 10 µA point sits below the datasheet's 90 %, because board leakage is counted.

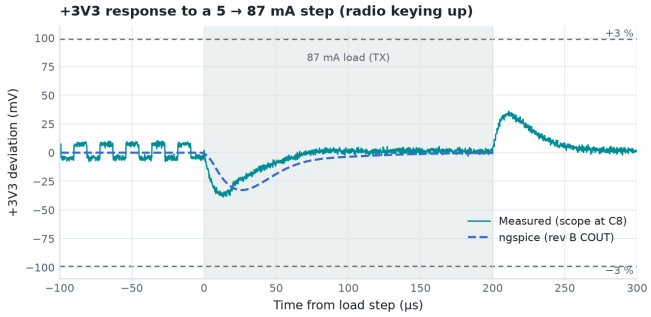


Figure 33. +3V3 under the 5→87 mA TX step: −40 / +35 mV (simulated −33 mV).

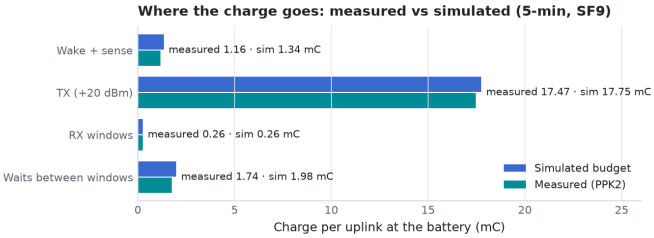


Figure 34. Charge per uplink by phase: measured against the simulation budget.

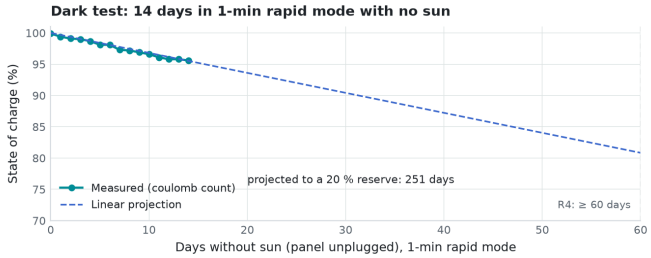


Figure 35. 14-day dark test in 1-minute mode: 0.32 %/day, so about 251 days to a 20 % reserve with no sun at all.

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9.2 Solar charging

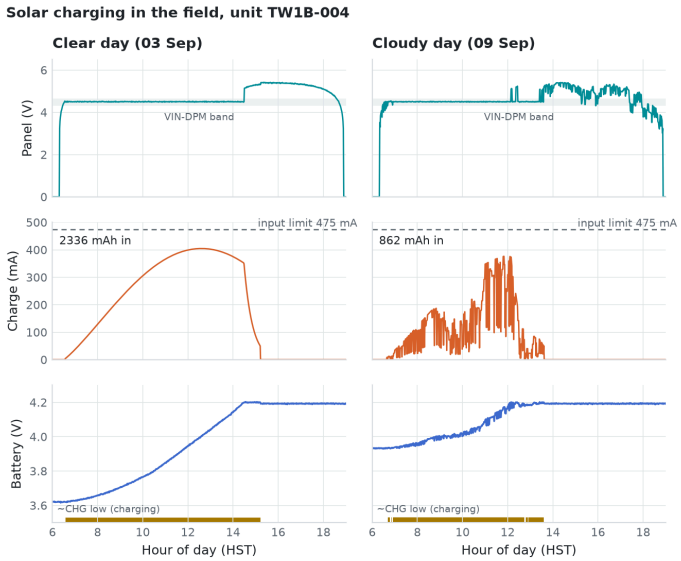


Figure 36. Field days on unit TW1B-004. Clear: 2,336 mAh in, panel held at 4.50 V while charging. Cloudy: 862 mAh. Daily use in 1-minute mode is about 9 mAh.

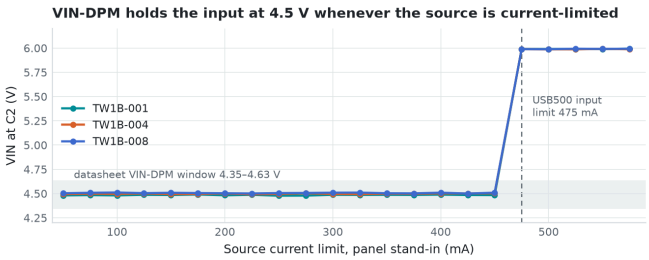


Figure 37. VIN-DPM sweep with an emulated panel: regulation at 4.485–4.506 V on 3 boards; input limit 474 mA.

Charge temperature window (NTC on the cell, chamber)

EDGE	CELL °C	EVENT
Rising	50.4	charging suspended
Falling	47.8	charging resumes
Falling	-0.6	suspended (cold)
Rising	2.1	resumes

Trips sit 0.4–0.6 °C outside the nominal 0 / 50 °C, within the BQ24074's TS threshold tolerance. Marked marginal against R6. In the field the cell never got above 37.1 °C, so this has no practical effect.

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9.3 Sensors

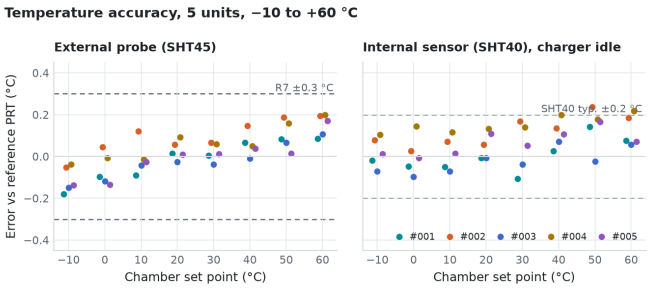


Figure 38. Air temperature, 5 units against a reference thermometer, −10 to +60 °C: worst 0.20 °C, RMS 0.10 °C (spec ±0.3).

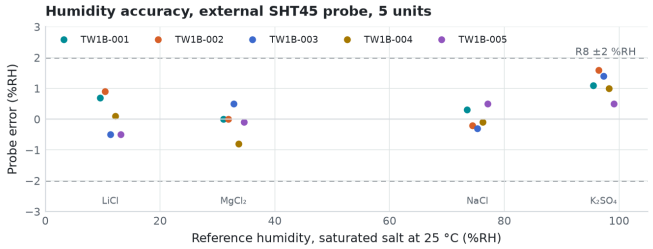


Figure 39. Humidity on saturated-salt references: worst 0.9 %RH below 90 %, 1.6 %RH at 97 % (spec ±2).

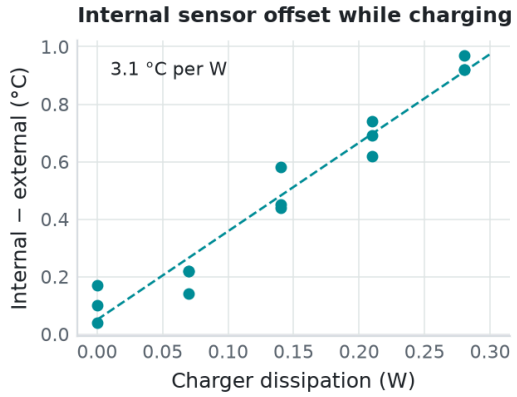


Figure 40. Internal SHT40 offset against charge current: 3.1 °C/W, 0.94 °C at 400 mA. Firmware removes it using the CHG status.

Rain counting (bench, tipping-bucket simulator)

RATE (TIPS/S)	APPLIED	COUNTED	MCU STATE
0.5	300	300	sleep (STOP2)
1.0	300	300	sleep (STOP2)
2.0	200	200	sleep (STOP2)
3.0	200	200	mixed (TX during burst)
5.0	100	100	sleep (STOP2)
10.0	100	99	sleep (STOP2)

Exact counting up to 5 tips/s, while a real gauge peaks near 2 tips/s in a 200 mm/h downpour. The single miss at 10 tips/s falls inside the 50 ms firmware lockout, which is there by design.

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9.4 Radio

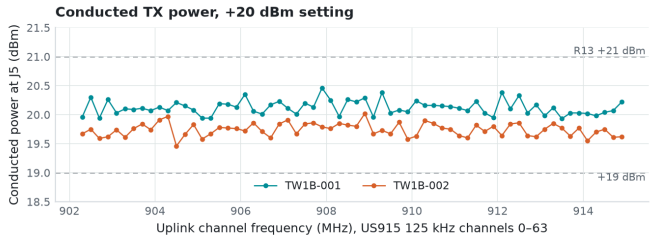


Figure 41. Conducted TX power at the U.FL, 64 channels × 2 units: 19.46 to 20.46 dBm, mean 19.93 dBm (spec +20 ±1).

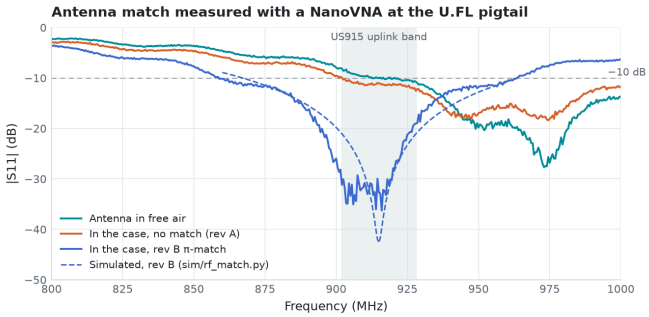


Figure 42. Return loss measured with a NanoVNA. Free air -10.0 dB; in the case without the match -11.1 dB; rev B match -32.7 dB at 915 MHz, -19.0 dB worst in band.

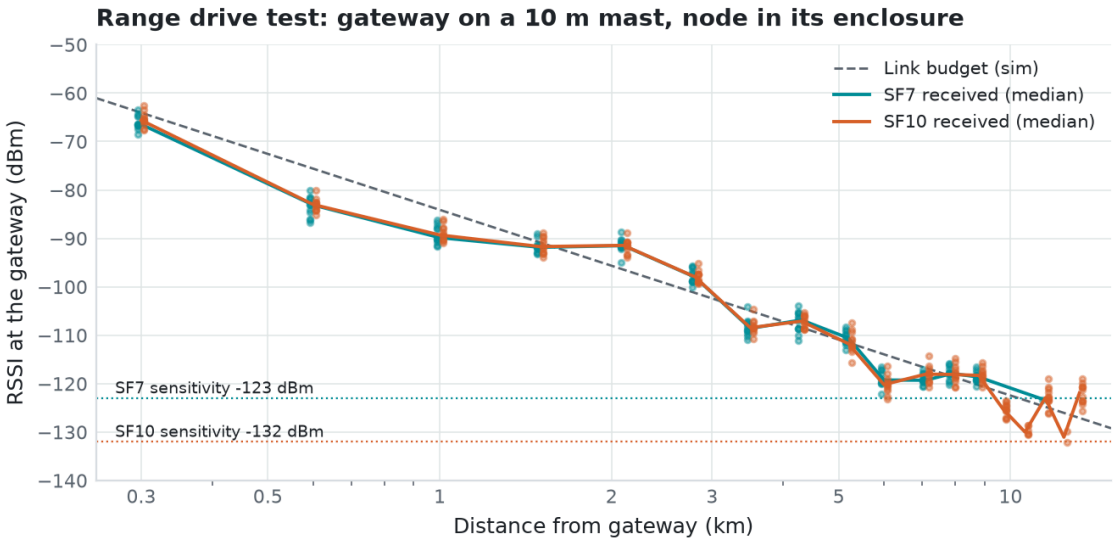


Figure 43. Drive test, gateway on a 10 m mast. SF7 held ≥ 50 % delivery to 8.8 km; SF10 still decoded at the last stop, 13.2 km, so its real limit lies further out.

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10 · ENVIRONMENTAL AND FIELD TESTING

10.1 Environmental

TEST	CONDITION	RESULT
Immersion	1 m, 30 min	PASS dew-point drift -0.05 °C
Hose	IPX5-style, 12.5 L/min, all faces	PASS
IP66 jet	100 L/min	OPEN needs accredited lab
Thermal cycling	−10/+60 °C, 50 cycles, 2 units	PASS gasket set 11 %
Operating range	−10 to +60 °C, chamber	PASS
Drop	26 × 1 m on concrete	PASS 2 mm cosmetic chip
UV	500 h UV-A, ASA coupons	PASS ΔE 1.7
ESD contact	±8 kV, all ports	PASS criterion A
ESD air	±15 kV	PASS B on J4 (1 reset, self-recovered)
Salt fog	ASTM B117	DEFERRED before coastal sites

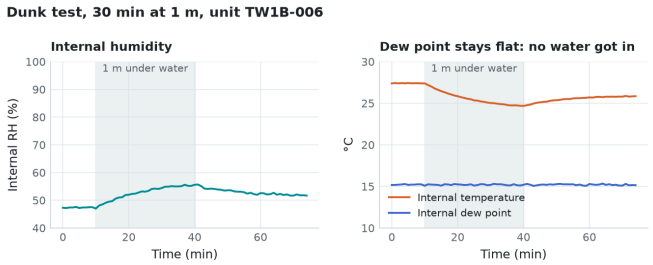


Figure 44. Dunk test: the internal SHT40 shows no humidity or dew-point change through 30 minutes at 1 m.

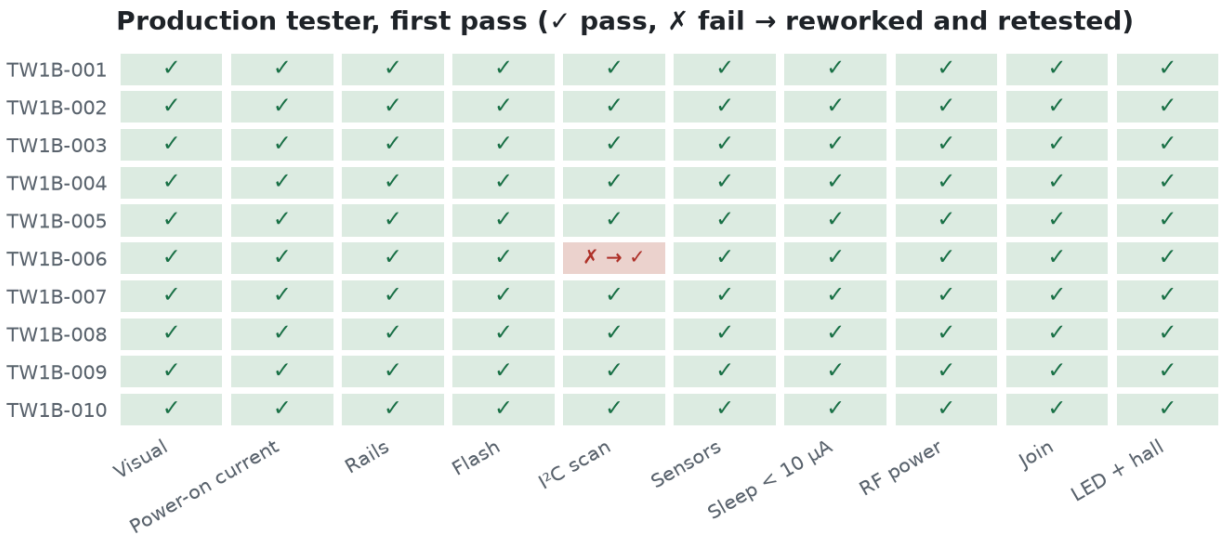


Figure 45. Production test, 10 boards × 10 steps: first-pass yield 90 %, final 100 % after one rework.

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10.2 30-day field pilot, O'ahu farm site, September 2026

Three units at 0.6, 1.4 and 3.2 km from one gateway, 5-minute reporting, rain gauge, soil probe and air probe fitted. Median RSSI -72 / -88 / -103 dBm.

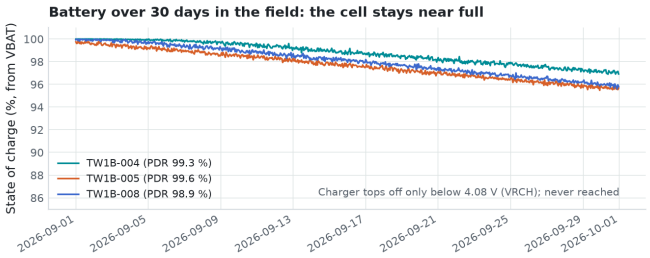


Figure 46. State of charge, 3 units at 0.6, 1.4 and 3.2 km: never below 94.6 %. Delivery 99.30 % (25,739 of 25,920 uplinks).

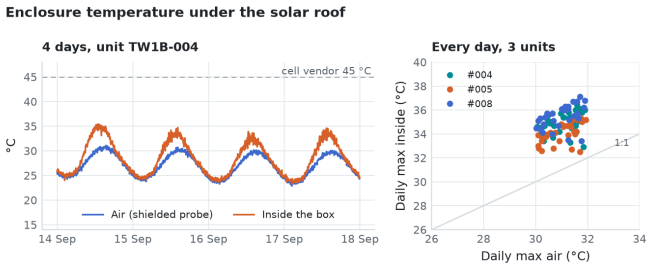


Figure 47. Box interior against air: max 37.1 °C at 31.9 °C air. On still afternoons the box ran 2–4 °C warmer than the thermal model; the margin to 45 °C still held.

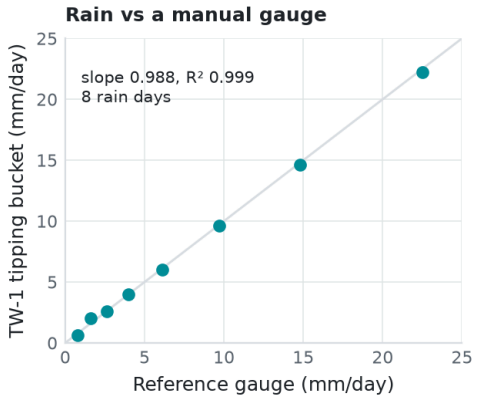


Figure 48. Rain against a manual reference gauge, 8 rain days: slope 0.988, R^2 0.9995.

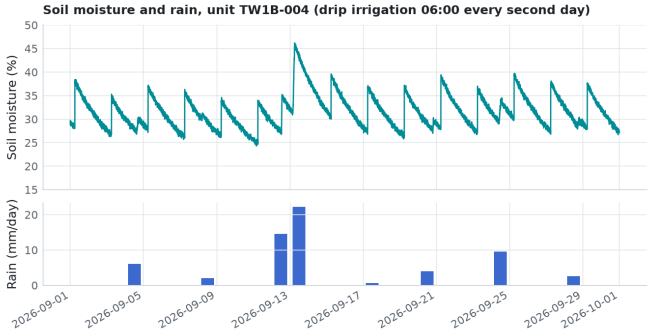


Figure 49. Soil moisture through irrigation cycles and rain events.

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11 · CONCLUSIONS AND NEXT STEPS

Rev B is ready for a pilot build. It meets every electrical, radio and sensing requirement with margin. Charging is marginal only at the edges of the NTC window. The enclosure's ingress rating needs one accredited IP66 jet test before it goes on a datasheet as certified.

Open items and risks

ITEM	RISK	PLAN
IP66 jet (R10)	Medium	Book an accredited lab; dunk and hose already pass
Charge window (R6)	Low	Optional Rs/Rp on TS to pull the hot trip to 45 °C (BQ24074 Table 9-3)
Internal sensor self-heating while charging	Low	Firmware compensation from the CHG status (0.94 °C at 400 mA)
FCC / IC	Low	Module is FCC-certified. Pre-scans: 11.6 dB margin radiated, 8.8 dB at the 3rd harmonic. Accredited §15.109 + host spurious test still to do
Salt fog (coastal sites)	Medium	ASTM B117 before any coastal deployment; consider conformal coat

Rev C changes (small)

- Layout review (4.4): move C5 and C2 next to U2, widen VBAT to 1 mm, C6 to 22 µF 0805.
- Open the U5 paste aperture 10 % to stop the pin 3–4 bridge.
- Larger radius on the lid's hinge-side corner (drop-test chip).
- Footprint for an optional TS trim network; silk polarity mark on the cell holder.

Recommended next step

1. **Pilot of 50 units**: about $3,400 for electronics, printed enclosures and kits, 4 weeks to assemble and test.
2. In parallel: book the IP66 jet and the accredited FCC run (about $2,500 together); the pre-scans say both should pass first time.
3. After 60 days in the field: freeze rev C and quote mold tooling for the box and lid.

Delivered with this report

KiCad 9 project (schematic, board, 3D models) · Gerbers, drill, BOM, pick-and-place · enclosure CAD (STEP, STL) and drawings · firmware and payload decoder with tests · simulation sources · test logs and analysis scripts · 3D renders · product datasheet. All in github.com/binaryBeachLLC/tradewind.

Maxwell Pauly

Maxwell Pauly · Binary Beach LLC · Hardware & product design

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